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WATTS NICKEL

THE COMPLETE TRAINING MANUAL

ENGINEERING NICKEL • DUCTILE • SULFUR-FREE

A full training course for the brand-new operator — from “what is an anode?” to a signed certificate of completion. Plain Watts nickel (engineering nickel) on steel, cast iron, and copper alloys. Assumes you know nothing. Teaches you everything.



OPERATOR TRAINING & CERTIFICATION

EDITION 1.0 • 2026 • BEGINNER-FRIENDLY STYLE

Training reference only. Always verify exact parameters against your chemistry supplier's Technical Data Sheets (TDS), customer specifications, current Safety Data Sheets (SDS), and applicable OSHA, EPA, REACH, and local regulations before production use. Nickel compounds are classified as suspected carcinogens and are known skin sensitizers. **Regulatory data reviewed 2026-09-17:** exposure limits reflect published U.S. federal OSHA PELs as of this date; referenced standards should be checked against their current revision, and state limits (e.g., Cal/OSHA) and ACGIH TLVs may be stricter.

— FIND YOUR WAY

TABLE OF CONTENTS

READ FRONT-TO-BACK THE FIRST TIME • KEEP ON THE SHELF FOREVER AFTER

PART ONE — FRONT MATTER & FUNDAMENTALS

Welcome Aboard & How to Use This Manual	3
Learning Objectives	4
Ch 1 • What Is Electroplating?	5
Ch 2 • What Is “Plain Watts Nickel,” and Why Do We Use It?	6
Ch 3 • How a Plating Line Works (The Big Picture)	9
Ch 4 • Why the Order Matters (Walking the Line)	10
Ch 5 • The Watts Bath — A Deep Dive	12
Ch 6 • Personal Protective Equipment (PPE)	18
Ch 7 • Emergency & First-Response	20
Ch 8 • The Safety Philosophy of a Nickel Line	22

PART TWO — THE PREP LINE (STATIONS 01-04)

Station 01 • Alkaline Soak Clean / Electroclean	24
Station 02 • Rinse (Pre-Activation): The Firewall	27
Station 03 • Acid Activation (The Acid Dip)	28
Station 04 • Rinse (Pre-Plate): The Guard Rinse	31

PART THREE — THE MAIN LINE (STATIONS 05-08)

Station 05 • Watts Nickel Plating — The Main Tank	33
Station 06 • Post-Plate Rinse & the Bake Decision	36
Station 07 • Post-Treatment (Bake, Passivate, Topcoat, or As-Plated)	37
Stage 08 • Final Rinse / Dry	38

PART FOUR — BACK MATTER & CERTIFICATION

Glossary — Every Term, Plain and Simple	39
Troubleshooting Quick-Index	41
Operator Training Matrix (Template)	42
Completion Test (20 Questions)	43
Answer Key	45
Certificate of Completion	46



REMEMBER

This manual is a training and reference tool — **not** a substitute for your facility's written procedures, your supplier's TDS/SDS, or your supervisor's instructions. When this manual and your shop's official procedure disagree, **follow your shop's official procedure and ask your supervisor.**

PART ONE — FRONT MATTER & FUNDAMENTALS

START HERE

WELCOME ABOARD

IF YOU DON'T KNOW AN ANODE FROM AN APRON, YOU ARE IN EXACTLY THE RIGHT PLACE

Welcome. If you are holding this manual, you are about to learn how to run — or at least understand — a **plain Watts nickel plating line**. Maybe you started this week. Maybe you have been racking parts for a year and finally want to know *why* you do the things you do. Either way, you are in the right place, and we are glad you are here.

Here is the most important thing we can tell you up front: **this manual assumes you know nothing about plating, chemistry, or electricity**. That is not an insult. That is a promise. Every term gets defined the first time we use it. Every step explains not just *what* to do but *why* it matters. Nobody is born knowing what “cathode efficiency” means or why a rinse tank can quietly ruin a whole shift. You learn it. We are going to teach it.

This manual is written in a **plain-language** style — friendly, plain-spoken, and patient. We would rather over-explain than leave you guessing in front of a tank full of hot acid nickel.

**REMEMBER**

You do not need to memorize this manual. You need to *understand* it. Understanding sticks. Memorizing fades by Friday.

• A WORD BEFORE WE START: THIS IS NOT BRIGHT NICKEL

You may have heard of — or worked on — a **bright nickel** line, where parts come out of the tank shiny as a mirror. **This is not that**. Plain Watts nickel is the *engineering* cousin of bright nickel. It uses almost the same core chemistry, but it leaves out the whole package of shine-making additives. The result comes out **matte to semi-lustrous** — a soft, satiny gray, not a mirror — and that is **exactly what we want**. A plain Watts deposit is ductile (it bends without cracking), sulfur-free, and built for *function*, not looks. Keep this in the back of your mind the whole way through: on this line, **matte is success, not failure**. We will come back to this idea again and again, because it changes how you read every part you pull.

• HOW TO USE THIS MANUAL

The manual follows the **line itself**, station by station, in the order a part actually travels: clean it, rinse it, activate it, rinse it, plate it, rinse it, post-treat it, and rinse/dry it. Reading it in order matters, because (as you'll soon learn) **the order of the plating line is not random** — and neither is the order of this book.

THE ICONS — YOUR FRIENDLY MARGIN HELPERS

As you read, you will run into little flagged notes. Here is what each one means.

**TIP**

A shortcut, a trick of the trade, or a smarter way to do something — the kind of thing a seasoned hand would lean over and tell you. These save you time and headaches.

**REMEMBER**

A core idea worth locking in. If you forget everything else on the page, don't forget these. They're the load-bearing ideas.

**HEADS-UP**

A safety warning. Read every single one. These are the notes that keep your skin, your eyes, and your lungs intact. We use them *liberally* because a nickel line has real hazards.

**TECHNICAL STUFF**

Deeper chemistry or theory for the curious. Totally optional. Skip these and you can still run the line perfectly. Read them and you will understand it on a deeper level.

**FUN FACT**

A bit of interesting trivia to make the science stick (and to give you something to say at lunch).

**TIP**

Read the Fundamentals all the way through *before* you read any individual stage chapter. The stage chapters assume you already understand the big picture this section gives you.

TWO MORE THINGS YOU'LL SEE AT EVERY STATION

Besides the margin icons, each station chapter ends with two recurring tools. The first is a **Teach-Back** checklist — the things you must be able to *say out loud, unprompted* before you're cleared on that station; it's how you and your trainer confirm the ideas actually stuck. The second is a bordered **Sign-Off** box at the very end of the station — the short statement your *trainer initials* (with you) once you've demonstrated the skill safely. Treat the Teach-Back as your study guide and the Sign-Off as the gate you have to pass through.

WHAT YOU’LL BE ABLE TO DO

LEARNING OBJECTIVES

THE WHOLE FOUNDATION — MASTER THIS AND EVERY STAGE CHAPTER CLICKS INTO PLACE

By the time you finish this manual, you should be able to confidently:

- 1. **Explain what electroplating is** in plain language — how electricity moves metal from one place onto another.
- 2. **Describe what plain Watts nickel plating does** and why a shop chooses it — a ductile, low-stress, sulfur-free, matte-to-semi-lustrous engineering deposit for wear, corrosion, build-up, and undercoat service.
- 3. **Explain how plain Watts differs from bright nickel** — same core bath, but no brightener system — and why you must never “fix” a matte deposit by adding brightener.
- 4. **Name all eight stages of the line in order** and explain *why* the order cannot be rearranged.
- 5. **Understand what each tank is for** — what goes in, what comes out, and what “good” looks like at every handoff.
- 6. **Recognize the main hazards** at each station — hot caustic, mineral acids, electrical, nickel mist and skin sensitization — and know the correct PPE and emergency response for each.
- 7. **Use the basic field tests** every operator relies on, like the water-break test, conductivity checks, and reading a Hull Cell panel for soundness (not brightness).
- 8. **Spot trouble early** — read a deposit and know whether the problem is upstream, in the bath, or in your handling.
- 9. **Talk the language** — use the right words so you can communicate clearly with your lead and your chemistry supplier.



REMEMBER

You are not expected to hit all of these on day one. Plating is a craft. The goal is steady competence, not instant mastery.

• THE EIGHT-STAGE PROCESS AT A GLANCE

Here is the whole plain Watts line in one strip. Don’t memorize it yet — just get a feel for the rhythm: **Clean → Rinse → Activate → Rinse → Plate → Rinse → Post-Treat → Rinse/Dry.**

#	STAGE	THE ONE THING IT DOES	TEMP	TIME
01	Alkaline Soak Clean / Electroclean	Removes all oil, grease, oxide, and shop soil	140–180°F	3–10 min
02	Rinse (Pre-Activation)	Washes off cleaner before the acid dip	Ambient	30–60 s
03	Acid Activation (Acid Dip)	Dissolves surface oxide; “wakes up” the metal	Amb–110°F	15–60 s
04	Rinse (Pre-Plate)	Guards the nickel bath from acid, iron & metals	Ambient	30–60 s
05	Watts Nickel Plating	Deposits the ductile engineering nickel layer	115–150°F	10–90+ min
06	Rinse (Post-Plate)	Washes off nickel drag-out; bake decision point	Ambient	30–60 s
07	Post-Treatment	Bake, passivate, topcoat, machine, or as-plated	Varies	Varies
08	Final Rinse / Dry	Cleans up and dries without water spots	Amb–140°F	30–60 s + dry



REMEMBER

Notice the pattern: **a rinse follows almost every working tank.** Rinses are not breaks — they are checkpoints that keep one chemistry from poisoning the next. On a nickel line this matters more than almost anywhere, because the nickel bath is *maintained rather than dumped* — a well-run bath runs for years. Every impurity you let ride into it on a part is a permanent guest.

CHAPTER 1

WHAT IS ELECTROPLATING?

THE TRUE BEGINNER'S PRIMER — SO FAR BACK WE DON'T EVEN ASSUME YOU KNOW WHAT PLATING IS

THE ONE-SENTENCE VERSION

Electroplating is using electricity to coat one metal with a thin layer of another metal. That's the whole idea. Now let's unpack it so it actually means something.

Imagine you have a steel shaft that is worn slightly undersize, or a bracket that needs to survive wear and corrosion in service. You want to give it a tightly-bonded skin of **nickel** — a tough, corrosion-resistant, semi-noble metal. You *could* try to spray or dip a coating on, but that would be uneven and weakly attached. Instead, we use electricity to grow a perfectly even, tightly-bonded layer of nickel directly onto the base metal, atom by atom. That process is electroplating.

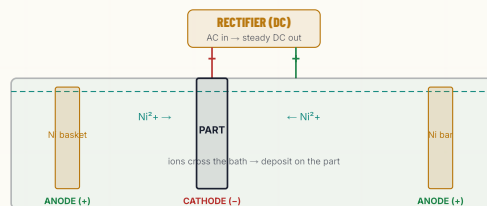
HOW ELECTRICITY MOVES METAL: THE KITCHEN-SINK PICTURE

Here is how it physically works. Picture a tank full of liquid called the **electrolyte** (or just “the bath”). This liquid has nickel dissolved in it — invisible nickel, broken down into tiny charged particles called **ions**. An *ion* is just an atom that carries an electrical charge. Dissolved nickel floats around the bath as positively-charged nickel ions, written **Ni²⁺**.

You hook two things up to a power supply called a **rectifier** (it converts ordinary AC wall power into the steady one-direction “DC” current plating needs):

- **The cathode** — this is your *part*, connected to the **negative** terminal. Just remember “cathode = the part, the thing collecting metal.”
- **The anode** — bars or baskets of nickel hanging in the same tank, connected to the **positive** terminal. As the bath uses up its dissolved nickel, the anodes slowly dissolve to replace it — they are the “fuel tank” of nickel metal. (“Anode = positive = the metal that feeds the bath.”)

When you turn on the rectifier: (1) current flows, and the positively-charged nickel ions (Ni²⁺) are *attracted* to the negatively-charged part, because opposite charges attract; (2) when a nickel ion reaches the part's surface, it grabs electrons and turns back into solid nickel metal, sticking to the surface; (3) meanwhile, at the anodes, solid nickel dissolves *into* the bath, releasing new Ni²⁺ ions to keep the supply topped up. The net result: nickel moves off the anodes, through the bath, and onto your part. The longer you plate and the more current you push, the thicker the coating.



REMEMBER

Three words to lock in now: **Cathode = the part** (collects nickel). **Anode = the nickel** (feeds the bath). **Electrolyte = the bath** (carries nickel from one to the other). Almost everything in plating is a variation on this one idea.



TECHNICAL STUFF

At the part, a **reduction**: $\text{Ni}^{2+} + 2\text{e}^- \rightarrow \text{Ni}^0$. At the anode, the reverse — solid nickel gives up two electrons and dissolves: $\text{Ni}^0 \rightarrow \text{Ni}^{2+} + 2\text{e}^-$. Because the anodes replenish what the cathode consumes, a well-run Watts bath is largely self-balancing on nickel metal — that is one reason it can run for years.



FUN FACT

The Watts bath is named for **Oliver P. Watts**, who characterized it in **1916**. More than a century later it is still the most widely used nickel electrolyte in the world. When you run this line, you are running chemistry older than television, radar, and the ballpoint pen — and it is still the industry workhorse.



HEADS-UP

Plating rectifiers push enormous current — hundreds to thousands of amps, low voltage but very high current, into a tank full of hot conductive liquid. **Never reach into an energized tank, and never do maintenance until equipment is locked and tagged out (LOTO).** Electricity plus hot salt solution plus your hand demands total respect.

CHAPTER 2

WHAT IS “PLAIN WATTS NICKEL”?

AND WHY DO WE USE IT – THE DUCTILE, LOW-STRESS, SULFUR-FREE ENGINEERING WORKHORSE

NICKEL PLATING COMES IN FAMILIES

“Nickel plating” just means coating something with nickel. But there are *several different recipes* for the bath that does it. The two you’ll hear about most are both built on the same Watts foundation:

- **Bright nickel** — a Watts bath *plus* a full package of organic additives (a carrier, a primary brightener, and a leveler). It comes out **mirror-bright** and is used where appearance matters. It is harder and less ductile, and it co-deposits a little sulfur.
- **Plain Watts nickel** — the *same* core bath **without** that additive package. It comes out **matte to semi-lustrous**, is **ductile** and **sulfur-free**, and is used for *engineering* purposes: wear, corrosion, build-up, undercoat, and electroforming. **This manual is about plain Watts nickel.**

★

REMEMBER – PLAIN WATTS IS NOT BRIGHT NICKEL

1. It is **NOT full mirror bright** — the deposit is matte to semi-lustrous, by design.
2. It **does NOT level** — it reproduces every scratch and tool mark in the substrate; surface prep must be right.
3. It has **NO brightener system** — no carrier, no primary brightener, no leveler. The only organic addition is the anti-pit wetter.
4. It **does NOT co-deposit sulfur** — the deposit stays safe for brazing, soldering, and thermal service.
5. It is **NOT brittle** — high ductility; it bends, forms, and machines without cracking.
6. It is **NOT a defective bright bath** — matte is the deliberate engineering choice: a stable, predictable, low-to-moderate tensile stress (set by temperature, pH, and current density), thick build-up, electroforming, and undercoats.

⚠

HEADS-UP – THE OPERATOR GUARD

Do NOT try to “fix” a matte deposit by adding brightener. A matte or satiny finish coming off a plain Watts bath is *correct*. If someone hands you a bottle of brightener and tells you to “shine it up,” stop and ask your lead — you would be turning an engineering bath into a different process entirely, changing the deposit’s ductility, sulfur content, and stress. Matte is the product here.

WHAT’S IN THE BATH

COMPONENT	TYPICAL RANGE	WHAT IT DOES
Nickel Sulfate (NiSO ₄ ·6H ₂ O)	225–300 g/L (30–40 oz/gal)	The main metal source — most of your nickel
Nickel Chloride (NiCl ₂ ·6H ₂ O)	30–60 g/L (4–8 oz/gal)	Helps anodes dissolve; carries current; improves throwing power
Boric Acid (H ₃ BO ₃)	30–45 g/L (4–6 oz/gal)	A pH “buffer” — keeps acidity steady right at the part’s surface; prevents burning
Wetting / anti-pit agent	Per supplier’s TDS (low level)	Prevents gas pitting — this is NOT a brightener
pH	3.5–4.5	Lowered with H ₂ SO ₄ , raised with nickel carbonate — never NaOH

⚙

TECHNICAL STUFF

“g/L” means **grams per liter** — how many grams of a substance are dissolved in each liter of bath. “oz/gal” is the same idea in ounces per gallon, which many US shops still use. A “**TDS**” is a **Technical Data Sheet** — the instruction document your chemistry supplier provides for each product. When this manual says “per TDS,” it means *follow the supplier’s sheet*. Never guess at additive levels.



REMEMBER — THE SINGLE ORGANIC INGREDIENT

The **only** organic addition to a plain Watts bath is the **wetting / anti-pit agent**, and its only job is to stop gas pits. There is no carrier, no brightener, no leveler. That absence is the whole reason this deposit is sulfur-free and ductile. It is the defining difference from a bright nickel line.



REMEMBER — BORIC ACID IS NOT OPTIONAL

It looks like a minor ingredient, but it is the buffer that holds the right acidity in the thin film of liquid right against the part where plating actually happens. Neglect the boric acid and you get **burning** at high-current areas even when everything else looks fine. Keep it above 30 g/L — and note that it dissolves slowly, so add it with heat and patience.

• WHY DO SHOPS USE PLAIN WATTS NICKEL?

Because it delivers a *functional* deposit with properties you can count on. Here is what it's good at:

APPLICATION	WHY PLAIN WATTS NICKEL
Engineering / functional coatings	Controlled thickness, predictable properties, good adhesion for wear and corrosion service
Corrosion protection	Dense, semi-noble deposit protects steel and copper alloys in industrial air
Undercoat / underplate	Barrier layer beneath chromium, gold, or other topcoats — and the ductile layer in duplex nickel systems
Salvage / build-up / restoration	Rebuilds undersized or worn parts to dimension — nickel can be deposited thick and machined back
Electroforming	Low-stress, sulfur-free deposit can be built thick and separated from a mandrel to make a freestanding part
Solderable / brazeable surfaces	Sulfur-free deposit does not embrittle or crack under downstream thermal joining
Diffusion / thermal-service parts	No co-deposited sulfur to migrate at grain boundaries when heated



FUN FACT

In **electroforming**, plain Watts nickel is grown *thick* onto a shaped “mandrel,” then the mandrel is pulled away — leaving a freestanding nickel part that is a perfect negative of the mold. Because the deposit is low-stress and ductile, it holds its shape and doesn't crack as it builds. Some music-record stampers, precision meshes, and mold tools are made exactly this way.

• THE DEPOSIT – WHAT “GOOD” LOOKS LIKE

PROPERTY	PLAIN WATTS (AS-PLATED, WELL-RUN)	BRIGHT NICKEL (TYPICAL)
Appearance	Matte to semi-lustrous (satiny)	Full mirror bright
Hardness	~150-250 HV (soft, ductile)	~350-550 HV (harder, more brittle)
Ductility	High — bends without cracking	Lower
Internal stress	Low-to-moderate tensile; stable and predictable from temp/pH/CD	Varies with additive balance; needs active control
Sulfur content	Essentially nil	Present (from the brightener carrier)
Thermal / braze suitability	Good — no sulfur to embrittle on heating	Poor — sulfur can embrittle at temperature

Hardness and stress figures are typical published ranges for plain Watts deposits and depend heavily on bath temperature, pH, and current density — treat them as engineering guideposts, not a spec guarantee. Always confirm against your supplier's TDS and the customer specification.

• THE HONEST TRADEOFFS

A good operator knows the weaknesses too:

- **No leveling.** Plain Watts reproduces every scratch, tool mark, and imperfection in the base metal. If the part looks rough going in, it looks rough coming out. Surface prep has to be right.
- **Not decorative as-plated.** Matte/semi-lustrous is not a mirror. If the customer wants shine, that's a topcoat or a buff — not this bath.
- **Sensitive to metallic contamination.** Copper, iron, zinc, and chromium sneaking into the bath cause dark, hazy, rough, or high-stress deposits. Contamination control is unchanged from bright nickel.
- **Runs hot.** Typically 115–150°F, so you pay to heat the tank (unlike a room-temperature process).
- **Hydrogen embrittlement risk** on high-strength steel — a bake is required (covered in detail later).



TIP

If a part comes off the line matte and someone calls it a “defect,” your first move is to explain the process — matte is correct for plain Watts. If they genuinely need shine, that's a *topcoat or buffing* decision made by your lead, not a bottle of brightener added to the tank.

CHAPTER 3

HOW A PLATING LINE WORKS

THE BIG PICTURE — A LINE IS A SYSTEM, NOT A PILE OF SEPARATE TANKS

A LINE IS A SEQUENCE OF TANKS

A plain Watts nickel line is not one tank. It's a **sequence of tanks** that a part travels through in a strict order. Think of it like a car wash: each station does one job, and each job prepares the surface for the next. Parts ride down the line in one of two ways:

- **Rack plating** — Parts are individually hung on a **rack** (a frame with hooks or clips that holds parts and carries current to them). Used for larger, heavier, or delicate parts, and for build-up/salvage where each part matters.
- **Barrel plating** — Lots of small parts are dumped together into a perforated rotating **barrel** that tumbles them through the chemistry. Efficient for high volumes of small parts, at lower current density.



TIP

You'll see different numbers for the same chemistry depending on rack vs. barrel — barrel baths often run a bit higher in nickel and chloride to make up for the lower current density tumbling parts receive. When in doubt about which numbers apply, check whether you're running rack or barrel and read the matching column on your shop-floor poster.

• THE WHOLE SEQUENCE: 8 STAGES

Here is the complete process. Don't memorize the details — just get a feel for the *flow*. Each stage gets its own full chapter later. The basic rhythm: **Clean** → **Rinse** → **Activate** → **Rinse** → **Plate** → **Rinse** → **Post-Treat** → **Rinse/Dry**.



• A LINE IS A SYSTEM, NOT A PILE OF SEPARATE TANKS



REMEMBER

Every stage either protects the next stage or poisons it. There is no neutral step. A part that leaves the cleaner still carrying a film of cleaner will contaminate the acid. A part that leaves the acid still wet with acid — and dissolved iron — will contaminate the nickel bath. What you do (or fail to do) at Stage 1 shows up as a defect at Stage 5. The line is a chain, and a chain is only as strong as its weakest link.

This is also why **the nickel bath itself is so precious**. Unlike the cleaner and the acid (dumped and remade regularly), a well-maintained Watts bath can run for *years*. But here's the catch: contaminants such as dissolved **iron, copper, and zinc** that sneak in tend to **stay in solution and accumulate**. You can't simply pour them out. Every milligram of junk you let drag into that tank is a permanent guest — you'll be dummy-plating and carbon-treating to chase it back out later.



TIP

Think of the nickel tank as the crown jewel of the line. Everything *before* it (clean, rinse, activate, rinse) exists largely to ensure only a perfectly clean, bare, well-rinsed part ever touches it. Everything *after* it (rinse, post-treat, rinse/dry) exists to protect and finish what the nickel created. The whole line orbits that one tank.

CHAPTER 4

WHY THE ORDER MATTERS

WALKING THE LINE — EACH STAGE SETS UP THE NEXT, AND THE SEQUENCE IS LOCKED BY CHEMISTRY

You might think the stage order is just convention — that you could shuffle steps or skip one to save time. You can't. Each stage exists *because of* the stage before and after it. Here is the logic, link by link.

STAGE 1 — CLEAN: GET RID OF THE JUNK FIRST

Plating, acid, and post-treatment all need to touch *bare metal*. Oil and shop soil block them. **Plate over dirt and you've just plated dirt** — the nickel bonds to grease, not metal, and peels off later. And remember: plain Watts has *no leveling and no brightener*, so it can't hide a prep failure. Every skipped clean plates through in plain sight as pits, skip plating, or peeling. There is a free, foolproof test for cleanliness called the **water-break test**: if water sheets off in an unbroken film, the part is clean; if it beads up like on a waxed car, oil remains.



HEADS-UP

The cleaner is **hot (140–180°F) and strongly alkaline (pH 11–13.5)** — it causes chemical burns and can seriously injure your eyes. Wear full PPE, and know where your shower and eyewash are before you need them.

STAGE 2 — RINSE (PRE-ACTIVATION): DON'T CARRY THE CLEANER FORWARD

The cleaner is alkaline (basic). The next tank is acid. If you carry cleaner straight into the acid, it neutralizes the acid, wastes it, and drags surfactants, silicates, and sodium downstream toward a bath where they don't belong. The rinse strips the cleaner film off first.



REMEMBER

"Drag cleaner into acid and you waste acid. Drag cleaner into the nickel bath and you poison a bath that never gets dumped." Rinses exist to keep each chemistry from invading the next tank. The cleaner-to-acid rinse is your first firewall on the line.

STAGE 3 — ACID ACTIVATION: WAKE UP THE METAL

Even a clean surface grows an invisible layer of **oxide** (a tarnish-like film from reacting with air). Nickel can't bond to oxide. The acid dip dissolves that oxide and leaves a chemically "fresh," active surface for the nickel to grip.

- **Under-dip** (too short/weak) = oxide remains = nickel peels off later (adhesion failure). On thick build-up and salvage work, this is fatal — the whole deposit is only as good as the bond beneath it.
- **Over-dip** (too long/strong/hot) = you start etching and pitting the part, and load it with hydrogen.



REMEMBER

No activation, no adhesion. Period. Activation has to come right before plating, with only a rinse in between — because freshly activated bare metal just grows a new oxide skin if it sits.



HEADS-UP

The acid dip uses **strong mineral acid** — typically HCl at 10–50% or H₂SO₄ at 5–20%. These cause severe burns and eye damage, and HCl gives off corrosive, choking fumes even at room temperature (OSHA PEL **5 ppm ceiling**). The acid eating into metal releases **flammable hydrogen gas**. Full acid PPE, good ventilation, and **always add acid to water, never water to acid**.

STAGE 4 – RINSE (PRE-PLATE): THE MOST IMPORTANT RINSE ON THE LINE

This is the **guard rinse** protecting your most valuable tank. Acid dragged into the nickel bath lowers its pH, and — critically — carries dissolved **iron** (and any copper or zinc from the pickle) straight into a bath that is maintained for years. Those metals accumulate and are hard to remove. This rinse is your primary contamination barrier, so it targets a tighter conductivity (**< 200 µS/cm**) and double rinses are strongly recommended.



REMEMBER

"Every contaminant you allow through this rinse becomes a permanent guest in the nickel." The pre-plate rinse is the single most important contamination barrier on the entire line. Drain well, rinse hard, and move the part quickly to the nickel tank before fresh oxide can form.



TECHNICAL STUFF

The recurring villain that justifies all those rinses is **drag-out** (and its mirror image, **drag-in**). Drag-out is the film of solution that clings to a part as it leaves a tank; carried into the next tank it becomes drag-in — contamination. Operators measure rinse cleanliness with **conductivity**, in microsiemens per centimeter (µS/cm) — basically how many dissolved ions are in the water. *Higher conductivity = more dragged-in chemistry = a dirtier rinse.* Critical rinses on this line target **under 200 µS/cm**.

STAGE 5 – WATTS NICKEL PLATING: THE MAIN EVENT

Now — and only now, with a perfectly clean, freshly activated, well-rinsed part — does the actual nickel plating happen. The bath is hot (115–150°F), acidic (pH 3.5–4.5), with the dissolving nickel anodes feeding fresh metal back into solution as you plate. It runs at high **cathode efficiency (93–97%)** — most of your current makes metal, not gas.

STAGE 6 – RINSE (POST-PLATE): RECOVER, AND DECIDE ON THE BAKE

The nickel bath is expensive and its drag-out contaminates any downstream tank. This rinse washes off the drag-out (a recovery rinse can capture nickel to return to the bath). It's also the **decision point for the hydrogen-embrittlement bake** on hardened steel.

STAGE 7 – POST-TREATMENT: FINISH FOR THE JOB

Plain Watts nickel is an *engineering* layer, so its "finish" depends entirely on the job: a **bake** (for hardened steel), an **anti-tarnish/passivate**, a **topcoat** (chrome, gold, lacquer), **machining to dimension** (for build-up/salvage), or nothing at all (as-plated undercoat or solderable surface).

STAGE 8 – FINAL RINSE / DRY: SEND IT OUT RIGHT

A last clean DI rinse and a warm (not scorching) dry finish the part without water spots or staining.



REMEMBER

The whole eight-stage sequence is one continuous logic chain: *clean it so nickel can stick → rinse so the cleaner doesn't ruin the acid → activate so the metal is bare and ready → rinse so acid and iron don't poison the precious nickel bath → plate the engineering nickel → rinse to recover nickel and decide on the bake → post-treat for the job → final rinse and gentle dry so it ships flawless.* Every step earns its place.



TIP

When you understand *why* each stage exists, troubleshooting becomes logical instead of mysterious. A peeling deposit? Look upstream at cleaning and activation. Cracking in a thick build-up? Look at bath contamination and current density. The defect almost always points back to the stage that was supposed to prepare for it.

CHAPTER 5

THE WATTS BATH: A DEEP DIVE

EVERYTHING HAPPENING IN THE CROWN-JEWEL TANK — CHEMISTRY, WINDOWS, ANODES, CONTAMINATION, STRESS, AND EMBRITTLEMENT

This chapter goes deeper than the others because the Watts bath is where the whole line pays off. Read it once now for the big picture; come back to it when you're troubleshooting.

• 5.1 — THE FOUR THINGS THAT ACTUALLY BUILD A DEPOSIT

On a bright nickel line, operators spend a lot of time babysitting brighteners. **You don't have a brightener system.** That frees you to focus on the four levers that actually control a plain Watts deposit:

1. **Metal** (nickel concentration) — how much nickel is available to plate.
2. **pH** — the acidity, which controls stress, burning, and efficiency.
3. **Temperature** — higher temp gives lower stress and higher ductility.
4. **Current density (CD)** — how hard you push; controls speed, and too much causes burning.



REMEMBER

"No brightener to chase. Just metal, pH, temperature, and current — the four things that actually build an engineering deposit." Master those four and you master this bath.

• 5.2 — OPERATING WINDOWS (RACK VS. BARREL)

PARAMETER	RACK	BARREL	NOTES
Temperature	115–150°F (46–66°C)	120–150°F	Higher temp = lower stress, higher ductility; target 130–140°F for engineering work
NiSO ₄ ·6H ₂ O	225–300 g/L	250–300 g/L	Higher for barrel to offset lower CD
NiCl ₂ ·6H ₂ O	30–60 g/L	45–60 g/L	Higher chloride improves anode dissolution in barrel
H ₃ BO ₃	30–45 g/L	37–45 g/L	Buffer — keep above 30 g/L; dissolves slowly
pH	3.5–4.5	3.5–4.5	Lower pH → lower stress but more H ₂ ; higher pH → risk of roughness/burning
Current Density	20–50 ASF (2–5.5 A/dm ²)	5–15 ASF	Higher CD = faster build but more burning/stress risk
Cathode Efficiency	93–97%	93–97%	Some H ₂ evolution always present
Deposition Rate	~0.001 in (25 μm) per hr at ~20 ASF	Lower	Scales with CD (~2 mil/hr at 40 ASF); verify by weight/coupon
Agitation	Air + cathode-rod movement	Barrel 6–12 RPM	Air agitation primary; it lowers pitting and evens thickness
Filtration	Continuous, 3–5 turnovers/hr	Continuous	5–10 μm filter bags; periodic carbon for organics
Anode:Cathode Ratio	2:1 to 1:1	Per basket config	Enough anode area to prevent passivation



TECHNICAL STUFF

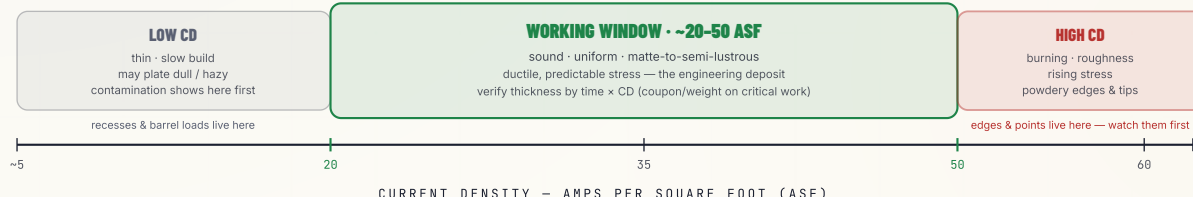
"ASF" = **amps per square foot**, the current density unit most US shops use ("A/dm²" is the metric version). Current density is current spread over the area you're plating. Sharp edges and points see *higher* local CD than flat faces — which is why edges "burn" first if you push too hard.

• THE CURRENT-DENSITY WINDOW: READ IT FOR SOUNDNESS

Because there is no brightener system, there is no “brightener window” to chase on this bath. What you manage instead is the **current-density window for a sound deposit**. Push too little current and the deposit builds slowly and can go dull or hazy at the low end; stay in the window and you get sound, uniform, matte-to-semi-lustrous engineering nickel; push too hard and the high-current areas burn, roughen, and stress up. The axis below is **sound deposit — not brightness**.

THE PLAIN WATTS CURRENT-DENSITY WINDOW — RACK WORK

the axis is a SOUND deposit, not brightness — there is no brightener to respond



• 5.3 — ANODE SELECTION: SULFUR-DEPOLARIZED VS. ROLLED/ELECTROLYTIC

This choice matters *more* for plain Watts than for bright nickel, because “sulfur-free” is one of your selling points.

- **Sulfur-depolarized nickel (S-Rounds / S-nickel):** contains a trace of sulfur that keeps the anode dissolving cleanly and efficiently, even at higher current densities. This is the best general-purpose choice — low anode polarization, forgiving to run.
- **Rolled depolarized / electrolytic (non-sulfur) nickel:** dissolves less readily and can **passivate** (go inert and stop dissolving) if anode area or chloride is marginal — but it introduces **no sulfur** into the system. Use it where the end use is genuinely sulfur-sensitive (electroforming for thermal service, brazing preforms, certain diffusion parts). If you run non-sulfur anodes, keep **chloride toward the high end (50–60 g/L)** and maintain generous anode area to prevent passivation.



TECHNICAL STUFF

A trace of sulfur *on the anode* is not the same as the sulfur *co-deposited into the deposit* by a bright-nickel brightener carrier. The “sulfur-free deposit” claim for plain Watts is about the absence of sulfur-bearing brighteners. Where a customer’s spec demands a truly sulfur-free system end-to-end, use non-sulfur anodes **and** no sulfur-bearing additives.



REMEMBER — ANODE BAGS

Always run **doubled polypropylene or cotton anode bags** to keep anode fines (little bits of undissolved anode) out of the bath — fines cause roughness. Leach new bags before first use.

• 5.4 – THE CONTAMINATION WATCH LIST

Because the bath is maintained for years, metallic contamination is your quiet, cumulative enemy. Know these action levels and what each contaminant does.

CONTAMINANT	ACTION LEVEL	EFFECT ON DEPOSIT	REMOVAL METHOD
Copper (Cu)	5 ppm	Dark streaks, roughness, poor low-CD deposit	Dummy plate at 2–5 ASF; carbon treat first
Iron (Fe)	25 ppm	Haze, roughness, raised stress, reduced ductility	Raise pH to 5.0–5.5 + H ₂ O ₂ to oxidize Fe ²⁺ →Fe ³⁺ ; filter; readjust pH
Zinc (Zn)	10 ppm	Hazy/dull, dark at low CD, adhesion loss	Dummy plate at 2–5 ASF; or raise pH to precipitate and filter
Chromium (Cr ⁶⁺)	1 ppm	Severe — dark/burnt, adhesion failure, skip plate	Reduce with sodium bisulfite to Cr ³⁺ , then treat as Cr ³⁺
Chromium (Cr ³⁺)	10 ppm	Dull, rough, poor throwing power and adhesion	Difficult — raise pH + dummy at low CD; or dilute
Lead (Pb)	0.5 ppm	Dark streaks, dendrites, stress cracking	Dummy plate at 2–5 ASF; check anode purity
Cadmium (Cd)	0.5 ppm	Peeling, blistering, dark deposits	Dummy plate; investigate source (legacy Cd drag-back)



REMEMBER – IRON HURTS YOU TWICE

In an engineering deposit, iron above ~25 ppm doesn't just cause haze and roughness — it can **raise internal stress and cut ductility**, undermining the very reason you're running a plain Watts bath. Keep the pickle fresh and the pre-plate rinse tight; iron drag-in is the most common way iron gets in.



HEADS-UP – CHROMIUM IS THE WORST OFFENDER

Even **1 ppm** of hexavalent chromium (Cr⁶⁺) wrecks adhesion and causes burnt, skip-plated deposits. The usual source is drag-back from a chrome line or a shared rinse. If a chrome process runs near your line, guard your rinses and racks fiercely.

• 5.5 – READING A HULL CELL (SOUNDNESS, NOT BRIGHTNESS)

A **Hull Cell** is a small angled test tank that plates a single panel across a whole *range* of current densities at once — high CD at one end, low CD at the other. On a bright nickel line, you’d read it for brightness and leveling. **On a plain Watts line, you read it for soundness:** burning, roughness, pitting, and contamination. Do **not** look for “brightness response” — there’s no brightener to respond.

PATTERN (267 ML, 2A, 10 MIN, ~130°F)	LIKELY CAUSE	CORRECTIVE ACTION
Sound, uniform matte/semi-lustrous across panel	Bath in balance	Continue monitoring metals, pH, boric acid
Burnt / powdery at high CD	pH too high; boric acid or Ni metal low; temp too low; CD too high	Check boric acid, pH, Ni metal; raise temp; lower CD
Rough / nodular (sandpaper feel)	Particulate — poor filtration; anode fines	Check filter pressure; inspect/replace anode bags
Gas pitting across panel	Low wetter; poor air agitation	Add anti-pit wetter; check air supply and sparger
Hazy / dull at low CD	Zinc, copper, or lead contamination	Analyze metals; dummy plate at 2–5 ASF
Dark streaks/spots at low CD	Copper or lead contamination	Analyze Cu and Pb; dummy plate
Peeling when panel is bent	Adhesion failure (prep) OR raised stress from metals/iron	Recheck cleaning/activation; analyze metals; carbon treat



TIP

Run the Hull Cell on a schedule (and after any correction), not just when something looks wrong. On a bath that is maintained for years, the panel is your early-warning radar — contamination shows up at the low-CD end of a panel long before it shows up on production parts.

• 5.6 – INTERNAL STRESS AND DUCTILITY

Every electrodeposit carries some **internal stress** — a built-in tension (or, less often, compression) locked into the metal as it grows. Too much *tensile* stress and a thick deposit can crack or peel. A big reason to run plain Watts is that, with no additives fighting you, its stress is **stable and predictable**, set by three things you control:

- **Temperature** — higher temperature lowers stress and raises ductility. This is why engineering work targets 130–140°F, not the low end.
- **pH** — lower pH tends to lower stress (but raises hydrogen evolution). It's a balance.
- **Current density** — pushing too high raises stress (and burning).

Keep contamination out (especially iron), run warm, and stay in the CD window, and you get the ductile, machinable, sound deposit plain Watts is known for.



TIP

If a thick build-up cracks during or after machining, don't reach for an additive. Look at **contamination (metals/iron), current density, and temperature** — the stress that cracked it was built in the tank. A sound plain Watts deposit machines well.

• 5.7 – HYDROGEN EMBRITTLEMENT (AND WHY A DUCTILE DEPOSIT STILL NEEDS THE BAKE)

Here is a subtle but life-critical idea. During cleaning, acid activation, and plating, some **atomic hydrogen** gets absorbed into the **steel substrate** — not the nickel. On high-strength/hardened steel (≥ 40 HRC / ≥ 180 ksi UTS), trapped hydrogen can cause the part to **crack suddenly under load**, sometimes hours or days after it's in service. On a critical fastener or structural part, that's a real safety failure.



HEADS-UP – A DUCTILE DEPOSIT DOES NOT EXEMPT THE PART FROM THE BAKE

People get confused here: “plain Watts is ductile and low-stress, so no bake, right?” **Wrong.** The ductility is a property of the *nickel*. The embrittlement is in the *steel underneath*, from hydrogen absorbed during processing. **High-strength steel still requires a post-plate bake.**

THE RULE (PER ASTM B850)

- Substrates ≥ 40 HRC (≥ 180 ksi UTS) require baking.
- Bake at $375 \pm 25^\circ\text{F}$ for a **minimum of 8 hours**.
- The clock starts at completion of the **last electrodeposition step**.
- **Bake within 4 hours** of plating — no exceptions.
- Some customer specs require baking *before* a topcoat plate — verify per spec.
- Where required, embrittlement is verified by test per **ASTM F519**.

• 5.8 — KEEPING THE BATH ALIVE: PURIFICATION

Because you never dump it, you *maintain* it. Two routines keep a Watts bath healthy for years:

CONTINUOUS PURIFICATION

Filtration (3–5 turnovers/hr through 5–10 µm bags), plus **dummy plating** (plating onto a scrap cathode at low CD, 2–5 ASF, to pull metallic contaminants out) as needed.

PERIODIC CARBON TREATMENT (TO REMOVE ORGANICS AND IRON)

1. Raise pH to 5.0–5.5 with nickel carbonate.
2. Add ~3% v/v H₂O₂ (30% grade) to oxidize Fe²⁺ → Fe³⁺.
3. Heat to 140–150°F.
4. Add 1–3 g/L powdered activated carbon; stir 2–4 hours.
5. Filter through a 1–5 µm filter to remove carbon and precipitated metals.
6. Readjust pH to 3.5–4.5 with dilute H₂SO₄.
7. Dummy plate at 2–5 ASF for 4–8 hours to remove residual metals.
8. Run a Hull Cell to confirm a sound deposit before resuming production.



FUN FACT

A plain Watts bath usually needs carbon treatment *less often* than a bright nickel bath — because there's no brightener continuously breaking down and loading the bath with organic byproducts. Fewer additives means fewer things to go wrong. That simplicity is part of why engineering shops like it.



HEADS-UP

Carbon treatment and pH adjustment mean handling **hydrogen peroxide, acid, and hot nickel solution** — and dummy plating means the rectifier is live. Full PPE, ventilation, and LOTO for any electrical work. This is skilled maintenance work; do it under supervision until you're signed off.

CHAPTER 6

PPE: YOUR DAILY ARMOR

THE GEAR THAT STANDS BETWEEN YOU AND CHEMICALS THAT BURN, MIST THAT HARMS, SPLASHES THAT BLIND

A nickel line works with hot caustic, strong mineral acids, electrical current at high amperage, chemical mists, and nickel chemistry that permanently sensitizes skin. None of these will hurt you if you respect them and wear the right gear. All of them can hurt you badly if you don't.

**REMEMBER**

There is no part, no order, and no deadline worth an injury. Scrap can be reworked. Your eyes — and your skin's tolerance for nickel — cannot.

• THE MASTER PPE SET (WEAR AT ALL STATIONS)

- **Chemical splash goggles** — sealed goggles, not safety glasses. Glasses leave gaps; a splash finds the gaps.
- **Face shield** — *over* the goggles for any work involving acids, hot caustic, or the plating bath.
- **Chemical-resistant gloves** — elbow-length, rated for acids and alkalis (nitrile, neoprene, or PVC). Inspect for pinholes before each use. **On a nickel line, gloves are non-negotiable at every station** — even dilute nickel sensitizes skin.
- **Chemical-resistant apron** — full-length, chest to below the knee.
- **Chemical-resistant boots** — closed, non-slip, splash-proof. Floors near plating tanks get wet and slick.
- **Long sleeves / appropriate clothing** — no exposed skin where splashes are likely.

**HEADS-UP – NICKEL IS A SKIN SENSITIZER**

Repeated skin contact with nickel — even weak, dilute solutions — can cause **allergic contact dermatitis, known as “nickel itch.”** Once you become sensitized, it is **cumulative and permanent** — you can react to nickel for the rest of your life, on the job and off. Wear gloves at *every* station where nickel is present, including the post-plate rinse where the solution is dilute. Do not let familiarity make you casual about bare-hand contact.

**HEADS-UP – NICKEL MIST AND DUST**

Air agitation in the plating tank throws up a fine nickel-containing mist. **OSHA PEL for soluble nickel compounds is 1 mg/m³ TWA (as Ni).** Nickel compounds are classified as suspected carcinogens. Local exhaust ventilation is mandatory over the tank; mist eliminators are recommended. When charging dry nickel salts or boric acid, avoid raising dust. Use the respirator (P100 or supplied-air) your shop specifies when ventilation is inadequate.

**HEADS-UP – BORIC ACID**

Handle boric acid as a **reproductive-hazard concern**. Avoid inhaling the dust when charging or making additions, wear gloves, and wash after handling.

**TIP**

Put your gear on *in order*: boots and apron first, then gloves, then goggles, then face shield last. Take it off in reverse. This keeps you from touching your face with contaminated gloves. Rinse your gloves *before* you take them off.

• STATION-BY-STATION HAZARD SUMMARY

STAGE	PRIMARY HAZARD	WHY IT'S DANGEROUS
01 Soak Clean / Electroclean	Hot alkaline (caustic), 140–180°F; H ₂ /O ₂ gas from electroclean	Chemical burns; hot splashes; flammable gas
02 Rinse	Residual alkaline cleaner	Mild skin/eye irritant; wet/slip hazard
03 Acid Activation	Strong mineral acid; HCl fumes (5 ppm ceiling); flammable H ₂ ; HF if die-cast option	Severe burns & eye damage; corrosive fumes; fire risk; HF is extreme
04 Rinse	Dilute acid residue	Mild irritant; slip hazard
05 Watts Nickel Plating	Hot acidic nickel bath; nickel mist (1 mg/m ³); high-amperage rectifier	Burns; skin sensitization; suspected carcinogen mist; shock/arc-flash
06 Rinse	Dilute nickel residue	Skin sensitizer even when dilute; slip hazard
07 Post-Treatment	Bake oven (burns/fire); anti-tarnish/seal; Cr(VI) if chrome topcoat	Burns; fire; Cr(VI) is a confirmed carcinogen
08 Final Rinse/Dry	Hot water/air; residual chemistry	Burns from hot dryer; slip hazard

**HEADS-UP**
Never eat, drink, smoke, vape, or apply cosmetics on the line. Hand-to-mouth contact is a major route for ingesting plating chemicals — especially nickel. Wash thoroughly before breaks and before leaving. Keep food and drink completely out of the work area.

CHAPTER 7

EMERGENCY & FIRST-RESPONSE

WHEN SECONDS COUNT — LEARN THIS COLD, BEFORE YOU EVER NEED IT

Know these *before* you need them. In an emergency, seconds matter and you won't have time to read. **Locate, before your first shift on the line:** the nearest **emergency eyewash** (within ~10 seconds' reach of every chemical station), the nearest **safety shower**, the **fire extinguisher** and alarm, the **rectifier emergency stop / disconnect** and main power shutoff, and the **spill kit** and **SDS binder** (Safety Data Sheets — the official hazard documents for every chemical on site).

**HEADS-UP**
Eyewash and shower stations must always be unblocked. Never stack parts, totes, or drums in front of them. The day you need one is the day you can't move boxes out of the way.

• FIRST RESPONSE BY HAZARD

SITUATION	WHAT TO DO — IMMEDIATELY
Chemical on skin	Get to the safety shower; flush at least 15 minutes . Remove contaminated clothing while flushing — don't stop early because "it feels better." Get medical attention; bring the SDS.
Chemical in eyes	Go to the eyewash; hold eyelids open and flush at least 15 minutes , rolling your eyes. Do not rub. Do not stop early. Get medical attention immediately — eye exposures are always urgent. Bring the SDS.
Inhaled fumes/mist, feel ill	Move to fresh air immediately. Report it — do not "tough it out." Seek medical attention for anything beyond brief, mild irritation. Bring the SDS.
Chemical swallowed	Do not induce vomiting unless the SDS directs it (vomiting acid or caustic burns the throat again). Follow SDS first aid; call medical help / poison control immediately.
Electrical shock / arc-flash	Do not touch the person if they may still be in contact with current. Cut power at the emergency stop / disconnect first, then call for help and render aid.
Chemical spill	Alert others and keep people away. Only clean it if trained and within your shop's "small spill" limits — otherwise evacuate and call trained spill response. Never let spills reach floor drains.

**HEADS-UP – HYDROFLUORIC ACID (HF)**

If your shop uses the fluoride activator for zinc die cast, know that **HF is uniquely dangerous** — it penetrates skin and attacks bone and can cause harm from exposures that don't hurt at first. A specific first-aid protocol and **calcium gluconate gel** must be on hand *before* HF is used. Any HF exposure is a medical emergency — flush and get help immediately.

**HEADS-UP – LOCKOUT/TAGOUT (LOTO)**

LOTO is mandatory for *all* rectifier and electrical maintenance. LOTO means physically locking the power off and tagging it so no one can switch it back on while someone is working. Never work on a rectifier that hasn't been locked out — "I'll just be a second" is how people get electrocuted.

**HEADS-UP – A CRITICAL ACID-MIXING RULE, FOR LIFE**

When diluting acid, **always add ACID to WATER — never water to acid**. Adding water to concentrated acid releases heat so violently it can boil and erupt the acid into your face. The memory aid: "**Do as you oughta — add acid to water.**" Pour slowly, down the side, with the acid going *into* a larger volume of water.

**FUN FACT**

The reason acid-to-water matters is heat capacity: a big pool of water absorbs the heat of dilution when the acid is the small addition. Get it backwards and the tiny bit of water flash-boils and spits acid. This one rule has saved more faces than any other in the chemical trades.

CHAPTER 8

THE SAFETY PHILOSOPHY

THE MINDSET THAT KEEPS YOU SAFE ACROSS A WHOLE CAREER

Rules and PPE are the *tools*. The philosophy is the *mindset* that makes them work. Five principles guide everything on this line.

1. ASSUME EVERY TANK CAN HURT YOU.

Even the rinse tanks carry residual chemistry — and on this line even a *dilute* nickel rinse can sensitize your skin. Even “just water” can be slippery enough to put you on the floor. Respect comes first; familiarity is where complacency — and accidents — begin.

2. CONTAINMENT BEATS CLEANUP.

Move parts deliberately. Let them drain over the tank before transferring. Don’t fling racks, splash, or rush handoffs. Most chemical exposures aren’t dramatic spills — they’re splashes and drips from sloppy handling. Slow, smooth movement is safer *and* makes better parts (less drag-out means cleaner rinses and a longer-lived nickel bath).

3. RESPECT THE TWO FACES OF EVERY HAZARD.

On this line, the things that make good parts are the same things that can hurt you. The acid that activates the metal burns your skin. The current that deposits nickel can arc-flash. The nickel that protects the part sensitizes your skin and its mist is a suspected carcinogen. You can’t separate the work from the hazard — so you manage both at once, every time.

4. THE BATH IS MAINTAINED, SO CONTAMINATION DISCIPLINE IS SAFETY DISCIPLINE.

Because the nickel bath runs for years, the same discipline that keeps it healthy — careful rinsing, no shortcuts, controlling drag-in, wearing gloves — is the discipline that keeps *you* healthy. Good housekeeping is a safety practice, not just a quality practice.

5. WHEN IN DOUBT, STOP AND ASK.

No one expects you to have every answer, especially while learning. The unsafe operator is not the one who asks questions — it’s the one who guesses. If a fume seems strong, a glove looks worn, a reading looks wrong, or a procedure feels off, *stop and check*.

• THE THREE HAZARD FAMILIES TO INTERNALIZE

- 1. **Acids & Alkali (corrosives).** The hot caustic cleaner (Stage 1) and the mineral acids (Stage 3) cause burns on contact and damage eyes instantly. The nickel bath itself is hot and acidic. *Defense:* goggles, face shield, gloves, apron — the 15-minute flush rule if exposed — and never forget acid-to-water.
- 2. **Electrical & physical.** The rectifier runs high amperage — shock and arc-flash risk. Wet floors mean slips. *Defense:* LOTO for all electrical work, never touch a rectifier or bus bar with wet hands, non-slip boots, keep walkways clear and dry.
- 3. **Nickel chemistry (sensitizer / suspected carcinogen).** Nickel sensitizes skin permanently (“nickel itch”) and its mist is a suspected carcinogen. *Defense:* gloves at every nickel station, local exhaust over the tank, respirator when ventilation is inadequate, and strict no-hand-to-mouth hygiene.

★

REMEMBER

Acids and alkali; electrical and physical; nickel chemistry. Three families. If you can name the hazard family at the tank in front of you, you already know most of what you need to protect yourself.

★

REMEMBER — YOU NOW HAVE THE WHOLE FOUNDATION

You’ve now got the whole foundation: what plating is, what makes plain Watts nickel special (and how it differs from bright nickel), how the eight-stage line works as one connected system, why the order is locked by chemistry, the bath in depth, what to wear, what to do in an emergency, and the safety mindset that ties it all together. Turn the page knowing the *why* — and the *how* will make a lot more sense.

ALKALINE SOAK CLEAN / ELECTROCLEAN

STRIP EVERY TRACE OF OIL, OXIDE, AND SHOP CONTAMINATION — NICKEL ADHESION IS BUILT HERE, NOT IN THE PLATING TANK

Welcome to the first working tank. Engineering nickel is often applied over machined, cast, ground, or previously plated parts that arrive carrying heavy oils and oxides — so cleaning is frequently the **most demanding step in the whole line**. And remember: plain Watts has no leveling and no brightener to hide a prep failure. A poorly cleaned surface plates as pits, skip plating, or peeling — *in plain sight*.

WHAT YOU'RE DOING & WHY IT MATTERS

Cleaning happens in two moves:

- **Soak clean** removes the *bulk* soil — oils, grease, shop grime — by soaking in a hot alkaline (caustic) cleaner.
- **Electroclean** scrubs off the *microscopic* film that soak-cleaning leaves behind, by running electrical current through the part in the cleaner. The gas bubbles it generates physically lift the last of the soil off the surface.

The whole point is a **water-break-free** surface: one where water sheets off in a continuous, unbroken film. If water beads up, oil remains.



REMEMBER — THE WATER-BREAK TEST IS FREE AND NON-NEGOTIABLE

Pull a cleaned part and watch how water sheets off it. A smooth, continuous, unbroken sheet (“zero seconds” to break) means the surface is clean. If water beads or “breaks” into droplets, there’s still oil there — **stop and re-clean**. It’s the simplest, most powerful quality check in the whole shop.

OPERATOR ESSENTIALS

PARAMETER	RANGE	NOTES
Soak Clean Temp	140–180°F (60–82°C)	Do not exceed 200°F — damages some cleaners
Soak Clean Time	3–10 min	Heavy draw oil: 8–10 min. Light stamping oil: 3–5 min
Soak Concentration	4–8 oz/gal (30–60 g/L)	Titrate weekly minimum
Electroclean Polarity	Cathodic → finish Anodic (last 15–30 s)	Anodic last strips smut back off
Electroclean Voltage	4–8 V	Current density 30–80 ASF typical
Electroclean Time	1–3 min	High-strength steel (>40 HRC): prefer anodic-only
Electroclean Temp	140–180°F	Same cleaner or dedicated electrocleaner
pH	11.0–13.5	Depends on formulation
Water Break Test	Pass = continuous sheet	Mandatory before proceeding

**TECHNICAL STUFF – CATHODIC VS. ANODIC CLEANING**

“Cathodic” means the part is negative (like during plating); it generates about *twice* the gas volume — more scrubbing — but can deposit metallic smut onto the surface. “Anodic” means the part is positive; less gas, but it strips that smut back off. That’s why the standard move is **cathodic first, then finish anodic for the last 15–30 seconds** — you get the scrubbing power, then clean up the smut.

**HEADS-UP – HIGH-STRENGTH STEEL: MINIMIZE CATHODIC TIME**

Cathodic cleaning loads hydrogen *into* the steel. On high-strength steel (>40 HRC), prefer **anodic-only** electrocleaning to limit hydrogen loading. (The post-plate bake is still required — this just keeps you from adding more hydrogen than you have to.)

**HEADS-UP – ZINC DIE CAST: GO GENTLE**

On zinc die castings, limit cathodic time to prevent hydrogen blistering of the casting, and use a mild cleaner at lower temperature — an aggressive hot caustic can etch and discolor die cast.

• STEP-BY-STEP PROCEDURE

1. **Confirm ventilation** and put on full PPE (goggles, face shield, elbow-length gloves, apron, boots).
2. **Verify temperature (140–180°F) and concentration** (titrate; 4–8 oz/gal). Skim any floating oil off the surface first.
3. **Immerse the rack fully** and soak for the part’s soil load (3–10 min).
4. **Electroclean:** run cathodic for the bulk of the time, then **finish anodic for the last 15–30 seconds**. For >40 HRC steel, run anodic-only.
5. **Pull and run the water-break test.** Continuous sheet = pass. Beading = re-clean; do not proceed.
6. **Transfer promptly** to the pre-activation rinse — a clean part left in air starts to re-film.

**HEADS-UP**

The cleaner is **hot (140–180°F) and strongly alkaline (pH 11–13.5)** — it causes chemical burns and can seriously injure your eyes. Electrocleaning evolves flammable hydrogen and oxygen gas at the surface. Full PPE, ventilation running, and know where your shower and eyewash are before you need them.

• TEACH-BACK CHECKLIST

- ☐ Explain what soak clean removes vs. what electroclean removes (bulk soil vs. microscopic film).
- ☐ Perform and interpret the water-break test (continuous sheet = clean; beading = oil remains).
- ☐ Explain why electroclean finishes anodic (to strip metallic smut back off).
- ☐ State when to use anodic-only (high-strength steel >40 HRC, to limit hydrogen loading).
- ☐ Explain why cleaning is unforgiving on a plain Watts line (no leveling/brightener to hide defects).

• TOP MISTAKES & QUICK TROUBLESHOOTING

SYMPTOM	PROBABLE CAUSE	WHAT TO DO
Water break fails	Low concentration or temp	Titrate cleaner; increase temp
Oily film after cleaning	Floating oil redepositing; exhausted cleaner	Skim/dump; recharge cleaner
White residue on parts	Hard water salts; heavy drag-out	Check water quality; improve rinse
Etching of zinc die cast	Cleaner too alkaline/hot for substrate	Reduce temp; use mild cleaner
Pitting in nickel downstream	Organic residue not removed	Add/extend electroclean; verify water break
Skip plating (bare spots)	Passive oxide film; fingerprints	Electroclean with anodic finish; handle with clean gloves
Peeling of thick build-up	Marginal cleaning under heavy deposit	Re-clean, re-activate, verify water break — prep is unforgiving for build-up

SIGN-OFF — STATION 01

Trainee: _____ Trainer: _____ Water-break: PASS / FAIL • Anodic-only used? Y / N

"I confirm I ran the soak clean at temperature and concentration, electrocleaned with an anodic finish (or anodic-only for high-strength steel), and verified a water-break-free surface before transferring."

STATION 02 OF 08

RINSE (PRE-ACTIVATION): THE FIREWALL

STRIP THE ALKALINE CLEANER FILM BEFORE THE ACID DIP – OR NEUTRALIZE IT IN THE WRONG TANK

You just cleaned the part in a hot, alkaline (basic) cleaner. The next tank is acidic. If you carry cleaner straight into the acid, it neutralizes and wastes the acid, and drags surfactants, silicates, and sodium downstream toward the nickel bath. This rinse is the **firewall** between the cleaning chemistry and everything after it.

OPERATOR ESSENTIALS

PARAMETER	RANGE	NOTES
Temperature	Ambient (60–85°F)	No heating required
Time	30–60 s with agitation	Longer for blind holes, tubing, recesses
Water Source	Municipal or DI	DI preferred for nickel lines
Conductivity	< 500 µS/cm (target < 300)	Monitor daily
Flow Rate	Continuous overflow or counter-current	2–4 gal/min typical
pH	7–10	Elevated pH = cleaner drag-in

★

REMEMBER – THE “CONTAMINATION NUMBER”
 Conductivity, in µS/cm, tells you how much dragged-in chemistry is in the rinse. Higher = dirtier. Here the target is **< 500 µS/cm**. If it creeps up, you’re dragging cleaner through — increase flow or add a rinse stage.

⚠

HEADS-UP
 The water carries residual alkaline cleaner. Wear goggles, chemical gloves, and non-slip footwear. Wet floors are a slip hazard.

TEACH-BACK CHECKLIST

- ☐ Explain why an alkaline cleaner and an acid must never meet on the part (neutralization wastes acid; drag-in pollutes downstream).
- ☐ State the conductivity target (< 500 µS/cm) and what a rising number means.
- ☐ Explain why blind holes and recesses need extra rinse time.

TOP MISTAKES & QUICK TROUBLESHOOTING

SYMPTOM	PROBABLE CAUSE	WHAT TO DO
High conductivity	Excessive drag-out; low flow	Increase drain time over cleaner; increase fresh water flow
Soapy residue on parts	Inadequate rinse time/volume	Extend dwell; add agitation; consider double rinse
Acid dip goes alkaline quickly	Cleaner carry-over overwhelming acid	Tighten rinse; add second rinse stage
Foaming in rinse	Surfactant from cleaner	Increase water flow; check cleaner formulation

SIGN-OFF – STATION 02

Trainee: _____ Trainer: _____ Conductivity: _____ µS/cm
“I confirm I rinsed to below the conductivity target with agitation, giving recesses extra time, before moving to the acid dip.”

STATION 03 OF 08

ACID ACTIVATION (THE ACID DIP)

DISSOLVE THE OXIDE LAYER AND EXPOSE BARE METAL FOR NICKEL ADHESION

Even a perfectly clean surface grows an invisible **oxide** film in air. Nickel can't bond to oxide — only to bare, chemically active metal. The acid dip dissolves that oxide. **Under-dip** and oxide remains, so the nickel peels. **Over-dip** and you etch the part and load it with hydrogen. For thick build-up and salvage work, adhesion is *everything* — the deposit is only as good as the bond beneath it.

• OPERATOR ESSENTIALS

PARAMETER	RANGE	NOTES
Temperature	Ambient-110°F	Ambient typical for mild activation
Time — Steel	15-60 s	Light oxide 15-30 s; heavy scale 30-60 s
Time — Brass/Copper	5-15 s	Short dip — acid attacks copper alloys fast
Time — Zinc Die Cast	5-15 s	Minimal; base metal dissolves rapidly; fluoride activation preferred
Acid (Option A)	HCl 10-50% v/v	Common; good oxide removal; fumes (5 ppm ceiling)
Acid (Option B)	H ₂ SO ₄ 5-20% v/v	Less fuming; slower; no chloride introduced
Acid (Option C)	10-20% H ₂ SO ₄ + 1-2% HF (die cast)	Activates die cast without excessive base-metal attack
Iron in bath	Dump when > 50 g/L Fe	Spent acid loses effectiveness; drags metals into nickel
Agitation	Manual rack movement	No air agitation — acid mist hazard

• SUBSTRATE-SPECIFIC NOTES

- **Steel / cast iron:** Standard HCl or H₂SO₄ activation. Cast iron (common in salvage/build-up) is porous — extend rinse and consider a **nickel strike** to seal the surface before heavy plating. Use **inhibited acid** for >40 HRC steel to reduce hydrogen absorption.
- **Brass / copper alloys:** Short dip only (5-15 s). Excessive time dissolves copper into the acid, which drags into the nickel bath.
- **Zinc die cast:** Extremely sensitive. Use a dilute fluoride-containing activator per supplier TDS — do **not** use strong HCl. After activation, die cast takes a **copper strike** (cyanide or alkaline non-cyanide) *before* Watts nickel — never directly into the acid nickel bath, or you get an immersion deposit, peeling, and zinc contamination.

• THE BIG ONE: HYDROGEN EMBRITTLEMENT



HEADS-UP – SAFETY & QUALITY: HYDROGEN EMBRITTLEMENT

When acid attacks steel, it releases hydrogen. Most bubbles away as gas — but some sneaks *into* the steel as single atoms, wedging into the crystal lattice. On ordinary mild steel this is usually harmless. But on **high-strength / hardened steel (≥ 40 HRC / ≥ 180 ksi UTS)**, trapped hydrogen makes the metal **brittle** — and a brittle part can **crack and fail suddenly under load**, sometimes hours or days later.

The acid dip is the single greatest source of hydrogen absorption on the entire line. Uninhibited acid on hard steel can load the lattice *in seconds*.

What you do about it: use the **minimum effective acid concentration and dip time** — don't over-dip "to be safe" (that's the opposite of safe here). **Always use an inhibitor on substrates over 40 HRC.** If a job is flagged high-strength, **stop and check with your lead** — and know that hardened parts require a post-plate **bake** (Station 06/07) per ASTM B850.

Terms: **HRC** = Rockwell C hardness (higher = harder). **ksi UTS** = thousands of psi of ultimate tensile strength. You don't calculate these — just treat them on a job traveler as a "slow down and check" flag.

• STEP-BY-STEP PROCEDURE

1. **Check the job for hardness flags first.** High-strength / hardened / ≥ 40 HRC / ≥ 180 ksi? Confirm the right process and that inhibitor is in use before anything else.
2. **Confirm ventilation is running** and your PPE (including face shield) is on.
3. **Verify acid strength and dip time** for this substrate (15–30 s clean steel; 30–60 s heavy oxide; 5–15 s copper alloys / die cast).
4. **Immerse fully and start your timer.** Agitate by gently moving the rack by hand — **never air**.
5. **Watch the surface.** Light bubbling is normal; the dark oxide should visibly lighten. If it isn't clearing, the acid may be dilute or spent — don't just leave a hard part in longer; check the bath.
6. **Pull at the timed mark.** Don't "round up." Over-dipping etches and embrittles.
7. **Drain briefly** and move **promptly** to the pre-plate rinse — activated metal re-oxidizes fast in air.

• TEACH-BACK CHECKLIST

- ☐ Explain what the acid dip removes and why nickel can't bond without it (oxide; no activation, no adhesion).
- ☐ Explain over-dip vs. under-dip consequences (etching + embrittlement vs. oxide remains + peeling).
- ☐ State dip times for clean steel vs. heavy oxide vs. copper alloys / die cast.
- ☐ Explain why there is NO air agitation here (acid mist hazard).
- ☐ Explain hydrogen embrittlement and which parts are at risk (≥ 40 HRC / ≥ 180 ksi), and the two things that reduce it (minimum dip, inhibitor).
- ☐ Recite the acid-dilution rule (always add acid to water).

• TOP MISTAKES & QUICK TROUBLESHOOTING

SYMPTOM	PROBABLE CAUSE	WHAT TO DO
Dark oxide remains	Acid too dilute; time too short	Increase concentration/time (<i>check embrittlement risk first</i>)
Pitting / etching	Too strong, too long, or too warm	Reduce concentration; shorten dip; add inhibitor
Adhesion failure / blistering under build-up	Incomplete activation; residual oxide; porous cast iron	Re-evaluate acid strength/time; consider nickel strike; check alkaline drag-in
Copper smut on brass	Dezincification — acid attacks zinc in alloy	Shorten dip; use copper strike before nickel
White smut on die cast	Fluoride too high or too long	Reduce concentration; shorten time; follow TDS
Excessive HCl fuming	High concentration/temp; poor ventilation	Reduce to minimum effective; check exhaust



HEADS-UP

Strong mineral acid — severe burns and eye damage. HCl fumes at room temperature (5 ppm ceiling). Hydrogen gas from pickling is flammable. If HF is used for die cast, it is extremely hazardous — calcium gluconate gel must be on hand. Full acid PPE and ventilation.

SIGN-OFF — STATION 03

Trainee: _____ Trainer: _____ Dip used: _____ s • Inhibitor? Y / N

"I confirm I checked the job for hardness flags, ran the acid dip at the timed interval with manual agitation only, ventilation and PPE in place, and used inhibitor where required."

STATION 04 OF 08

RINSE (PRE-PLATE): THE GUARD RINSE

THE LAST BARRIER BEFORE YOUR NICKEL BATH — A TANK THAT MAY RUN FOR YEARS WITHOUT A DUMP

You're at the last station before the nickel plating bath — the crown jewel of the line. Your job here is to **wash the acid (and the dissolved iron and metals it carries) off the part** so it doesn't contaminate that expensive, maintained-for-years bath. Think of this rinse as the **guard at the door** of the most valuable room in the building.

WHAT YOU'RE DOING & WHY IT MATTERS

Coming out of the acid dip, the part is wet with acid — and that acid now carries **dissolved iron** (and possibly copper or zinc) it picked up while pickling. Carry that into the nickel bath and you cause three problems:

- 1. **You drop the bath's pH.** The Watts bath runs in a tight window (3.5–4.5); extra acid pushes pH down, changing stress and risking burning.
- 2. **You add iron.** Iron accumulates in the bath, causing haze, roughness, raised stress, and reduced ductility — and it's hard to remove.
- 3. **You add other metals** (Cu, Zn) that accumulate the same way.

★

REMEMBER — EVERY CONTAMINANT HERE IS A PERMANENT GUEST
Because the nickel bath is maintained rather than replaced, every milligram of acid, iron, copper, or zinc that enters through poor rinsing accumulates. This is the single most important contamination barrier on the line. Hold it to a tighter standard than any rinse before it.

OPERATOR ESSENTIALS

PARAMETER	RANGE	NOTES
Temperature	Ambient (60-85°F)	No heating required
Time	30-60 s with agitation	Double rinse strongly recommended
Water Source	DI strongly preferred	Municipal water adds calcium, chloride, more
Conductivity	< 200 µS/cm (target < 100)	Critical for nickel bath life
pH	4.0-8.0	Below 4 = excessive acid drag-in
Flow Rate	Continuous overflow, 3-5 gal/min	Counter-current ideal
Drain Time	15-20 s minimum	Prevents diluting the nickel bath



REMEMBER — THE 30-SECOND CLOCK

Move into the nickel bath **within 30 seconds** of leaving the rinse. A freshly activated, rinsed surface starts to oxidize in air almost immediately, and an oxidized surface plates poorly — undoing Station 3's work. Keep the part wet; speed matters at this handoff.



HEADS-UP

The water carries dragged-in acid — diluted but still irritating. Goggles, chemical gloves, non-slip footwear. Parts must move quickly to the nickel tank — no delays.

• **TEACH-BACK CHECKLIST**

- ☐ Explain the three ways acid/iron drag-in hurts the nickel bath (pH drop, iron, other metals).
- ☐ State the conductivity target (< 200 µS/cm) and why it's tighter than Station 02.
- ☐ Explain the 30-second rule and why re-oxidation matters.
- ☐ Explain why double-rinsing is recommended here specifically.

• **TOP MISTAKES & QUICK TROUBLESHOOTING**

SYMPTOM	PROBABLE CAUSE	WHAT TO DO
Conductivity high	Heavy acid drag-out; low flow	Increase drain time from acid dip; increase flow
Nickel bath pH dropping	Acid drag-in from activation	Tighten conductivity target; add second rinse
Iron climbing in nickel bath	Dissolved Fe from spent pickle	Refresh pickle; improve rinse; add a stage
Parts oxidizing before plating	Transfer too slow; parts drying	Speed transfer — enter nickel within 30 s
Chloride climbing in nickel bath	HCl drag-in	Switch to H ₂ SO ₄ activation; or improve rinse

SIGN-OFF — STATION 04

Trainee: _____ Trainer: _____ Conductivity: _____ µS/cm • To-tank time under 30 s? Y / N

"I confirm I rinsed below the tight conductivity target (double rinse where available), drained well, and moved the part to the nickel bath within 30 seconds."



REMEMBER — THE PREP-LINE PROMISE

Four stations, one mission: hand the nickel bath a surface that is **clean, contaminant-free, oxide-free, and still wet**. Clean it (01), rinse the cleaner off (02), strip the oxide (03), rinse the acid and iron off (04), and move fast. Every one a checkpoint, not a rest stop. Do these four right, every rack — and sound engineering nickel will reward you. Cut corners here, and no amount of bath chemistry downstream can save the job.

PART THREE — THE MAIN LINE (STATIONS 05–08)

STATION 05 OF 08 • THE MAIN TANK

WATTS NICKEL PLATING

NICKEL SULFATE, NICKEL CHLORIDE, BORIC ACID, AN ANTI-PIT WETTER, AND NOTHING ELSE — DUCTILE, LOW-STRESS, SULFUR-FREE BY DESIGN

This is the payoff. With a perfectly clean, freshly activated, well-rinsed part, the engineering nickel goes on. You already met this bath in depth in Chapter 5 — this chapter is how you *run* it, shift to shift.

**REMEMBER — WHAT YOU'RE BUILDING**

A **matte to semi-lustrous, ductile, sulfur-free** deposit. If it comes out satiny gray, that's *success*. You are not chasing shine — you are building sound, controlled-thickness engineering nickel. *"No brightener to chase. Just metal, pH, temperature, and current."*

• OPERATOR ESSENTIALS (SEE CH 5 FOR THE FULL WINDOW TABLE)

- **Temperature:** 115–150°F (target 130–140°F for engineering work — warmer = lower stress, higher ductility).
- **pH:** 3.5–4.5. Lower with H_2SO_4 , raise with nickel carbonate — **never NaOH**.
- **Current density:** 20–50 ASF rack; 5–15 ASF barrel. Stay in the window — too high burns edges and raises stress.
- **Boric acid:** keep above 30 g/L (it buffers against burning).
- **Air agitation + filtration:** both running continuously — agitation lowers pitting and evens thickness; filtration keeps particulate out.
- **Anodes:** S-Rounds for general work; non-sulfur only when the end use is sulfur-sensitive (then keep chloride high). Doubled anode bags always.

• THE RULE CARD: 93–97% CATHODE EFFICIENCY

**TECHNICAL STUFF**

Cathode efficiency means what fraction of your current actually deposits metal. The Watts bath runs at **93–97%** — most of your current makes nickel, not gas. But the small remainder still evolves hydrogen, and enough atomic hydrogen loads into high-strength steel to embrittle it. The *deposit* is ductile; the *substrate* still needs the bake.

• STEP-BY-STEP PROCEDURE

1. **Confirm PPE and ventilation.** Gloves at all times (nickel sensitizer); local exhaust on (nickel mist).
2. **Check temperature, pH, and boric acid** at shift start; confirm agitation and filtration are running.
3. **Confirm the CD / plating time** for the target thickness on the job ticket. Enter the tank within 30 s of the pre-plate rinse.
4. **Immerse, connect to the cathode bar, and start.** Watch for even gassing and no arcing at contacts.
5. **Plate to the target thickness** (time $\times$ CD; verify by coupon/weight for critical work — deposition $\sim 25 \mu\text{m/hr}$ at ~ 20 ASF).
6. **Pull, drain over the tank** (recover drag-out), and move to the post-plate rinse.
7. **Log the run.** Note anything off — burning, pitting, roughness — for the next Hull Cell / maintenance check.

• READING TROUBLE (SEE CH 5.5 FOR THE FULL HULL CELL GUIDE)

SYMPTOM	PROBABLE CAUSE	WHAT TO DO
Burning at high CD	pH too high; boric acid or Ni metal low; temp too low; CD too high	Check pH, boric acid, Ni metal, temp; lower CD
Dull/hazy deposit	Metallic contamination; organic drag-in; low temp	Analyze metals; carbon treat; check temp
Pitting (gas pits)	Low anti-pit wetter; poor agitation; organics	Add wetter; increase air; carbon treat; check filter
Roughness / nodules	Particulate; poor filtration; anode fines	Check filters; replace anode bags; raise filtration
Peeling / poor adhesion	Poor cleaning; incomplete activation; oxidized surface	Recheck water break; recheck acid dip; speed transfer
Cracking in thick build-up	Raised stress; iron/metal contamination; CD too high	Analyze metals; carbon treat; lower CD; raise temp
Poor throwing power	Low NiCl ₂ ; pH too low; low temp; poor agitation	Raise chloride to 50–60 g/L; optimize pH/temp; improve agitation
Anodes passivating (voltage climbing)	Low chloride; too little anode area; non-sulfur anodes at high CD	Raise chloride; add anode area; switch to S-nickel if sulfur is acceptable
Deposit too soft / building too slow	CD too low; low metal; efficiency loss from contamination	Raise CD within window; check metal; analyze contaminants



HEADS-UP – THE MAIN-TANK SAFETY CARD

Hot acidic nickel solution (pH 3.5–4.5, up to 150°F) — causes burns and is a **skin sensitizer**; nickel compounds are **suspected carcinogens**. “Nickel itch” is cumulative and permanent once sensitized — minimize skin contact always. Air agitation generates **nickel mist — local exhaust mandatory (PEL 1 mg/m³ as Ni)**. Rectifier = high amperage — **LOTO for all maintenance**. Handle **boric acid** per its reproductive-hazard note (no dust, gloves, wash after). PPE: goggles + face shield, elbow-length nitrile gloves, apron, boots, P100 or supplied-air respirator if ventilation is inadequate.



HEADS-UP – TANK MATERIAL

This is a hot, acidic, chloride-containing bath. Tanks, heaters, and fixtures must be **PP, CPVC, or lined** — **never bare mild steel**, which the bath would corrode (dumping iron into your bath and failing structurally). Plain PVC is marginal above 140°F, so CPVC is the safer plastic for this hot bath.

• TEACH-BACK CHECKLIST

- ☐ Name the four control levers (metal, pH, temperature, current density) and what each does.
- ☐ Explain why matte is correct and why you never add brightener to fix it.
- ☐ State the pH range and the correct way to raise/lower it (nickel carbonate up, H_2SO_4 down; never NaOH).
- ☐ Explain why the deposit is ductile but the steel still needs the bake.
- ☐ State the nickel PEL (1 mg/m^3 as Ni) and the sensitizer / suspected-carcinogen hazards.
- ☐ Explain why the tank must be PP/CPVC/lined, not mild steel.

SIGN-OFF – STATION 05

Trainee: _____ Trainer: _____ Temp: _____ °F · pH: _____ · CD: _____ ASF ·
 Target thickness: _____ μm

"I confirm I ran the bath in its temperature/pH/CD window with agitation and filtration, plated to target thickness, wore gloves with exhaust running, and did not attempt to brighten a matte deposit."



TIP

Keep a simple tank log at this station — temperature, pH, boric acid, and anything unusual, every shift. On a bath that runs for years, the log is how you see slow drift before it becomes a defect. The best operators can tell you what their bath “normally” reads without looking it up.

STATION 06 OF 08

POST-PLATE RINSE & THE BAKE DECISION

WASH OFF THE DRAG-OUT BEFORE POST-TREATMENT — AND DECIDE WHETHER TO BAKE FIRST

Fresh out of the nickel tank, the part carries **nickel drag-out** — nickel sulfate and chloride clinging to the surface. That drag-out wastes nickel and contaminates any downstream tank (passivate, seal, topcoat, or a second plating bath). This rinse washes it off — and a **drag-out recovery rinse** (a still first rinse) can capture nickel to return to the bath. This is quality control and cost control at once.

• OPERATOR ESSENTIALS

PARAMETER	RANGE	NOTES
Temperature	Ambient	No heating required
Time	30–60 s with agitation	Double rinse recommended
Conductivity	< 200 µS/cm (target < 100)	Nickel drag-out is the concern
Drain Time	15–20 s over nickel tank	Maximizes drag-back recovery
Drag-Out Recovery	Optional still first rinse	Periodically return to bath to recover Ni

• THE BAKE DECISION POINT



HEADS-UP — HYDROGEN-EMBRITTLEMENT BAKE

For hardened steel (> 40 HRC): **bake within 4 hours of the last electroplating step at 375 ±25°F for a minimum of 8 hours**, per ASTM B850. The 4-hour clock starts at completion of the last electrodeposition step. Verify your customer spec — some require baking *before* any post-treatment or second plating layer. If baking is required, parts go from this rinse to the oven.



REMEMBER

The plain Watts *deposit* is ductile and low-stress — but the hydrogen absorbed by the *substrate* during cleaning, activation, and plating is the embrittlement concern, not the nickel. **A ductile deposit does not exempt a high-strength steel part from the bake.**

• TEACH-BACK CHECKLIST

- ☐ Explain why post-plate rinsing protects both the nickel budget and the next tank.
- ☐ State the bake rule for >40 HRC steel (375 ±25°F, ≥8 hr, within 4 hr, per ASTM B850) and where the clock starts.
- ☐ Explain why a ductile deposit doesn't remove the bake requirement.

• TOP MISTAKES & QUICK TROUBLESHOOTING

SYMPTOM	PROBABLE CAUSE	WHAT TO DO
Downstream tank life short	Nickel drag-in	Tighten this rinse; add recovery rinse
Water spots on nickel	Hard water; slow transfer	Use DI; minimize air exposure
Haze / staining on nickel	Oxidation during slow transfer; drag-out drying	Reduce transfer time; keep parts wet
Nickel salt crystallizing on rack	Inadequate rinsing; high drag-out	Extend rinse; strip racks more often

SIGN-OFF — STATION 06

Trainee: _____ Trainer: _____ Conductivity: _____ µS/cm · Bake required? Y / N · If Y, oven within 4 hr? Y / N

"I confirm I rinsed off nickel drag-out to target and correctly identified whether the part required the H-embrittlement bake, routing hardened parts to the oven within the 4-hour window."

STATION 07 OF 08

POST-TREATMENT

BAKE, PASSIVATE, TOPCOAT, OR TAKE IT AS-PLATED — ENGINEERING NICKEL’S FINISH IS DEFINED BY ITS END USE, NOT BY APPEARANCE

Here is where plain Watts differs sharply from bright nickel. A bright nickel line almost always ends in decorative chrome. Plain Watts is *engineering* nickel — its “post-treatment” is more often a **bake**, a **topcoat**, **machining to dimension**, or **nothing at all**. What you do is dictated by the job, not by appearance.

• POST-TREATMENT OPTIONS

TREATMENT	TYPICAL USE	KEY PARAMETER
H-embrittlement relief bake	High-strength steel (>40 HRC)	375 ±25°F, ≥8 hr, within 4 hr of plating (ASTM B850)
As-plated (no treatment)	Undercoat before topcoat; internal parts; solderable/brazeable surfaces	Store with VCI/desiccant to limit tarnish; keep clean
Anti-tarnish / passivation dip	Delay tarnish on exposed nickel	Per supplier TDS — a mild inhibitor dip, not electrodeposited
Chromate / conversion seal	Added corrosion resistance without a topcoat plate	Per supplier TDS
Clear lacquer / topcoat	Indoor decorative or handling protection	5–15 µm; air or heat cure per TDS
Chromium topcoat (decorative or hard)	When wear or appearance demands it over the nickel	Decorative Cr 0.25–0.75 µm; hard Cr per its own process
Gold / precious topcoat	Electronics, connectors	Nickel serves as a diffusion-barrier underplate
Machining / grinding to dimension	Salvage / build-up / restoration	Deposit built oversize, then machined to final size

• AS-PLATED & STORAGE

Plain Watts nickel is corrosion-resistant but **not** corrosion-proof — it slowly tarnishes/hazes in humid or industrial air. For parts stored before assembly or topcoating, keep them clean, dry, and protected (VCI packaging or desiccant). For a subsequent plating layer (chrome, gold), **re-activate** the nickel — it passivates on standing and needs a light acid dip to restore adhesion before the next deposit.

★

REMEMBER — RE-ACTIVATE BEFORE A TOPCOAT

Nickel that has sat around grows a thin passive film. If you plate chrome or gold onto it without a light re-activation dip, the topcoat can peel. Minimize the delay between plating the nickel and applying the topcoat.

• IF A CHROMIUM TOPCOAT IS APPLIED



HEADS-UP – HEXAVALENT CHROMIUM

If the job calls for decorative or hard chromium over the nickel, that chrome process carries its own severe hazards. **Hexavalent chromium (Cr⁶⁺) is a confirmed human carcinogen (IARC Group 1). OSHA PEL: 5 µg/m³ TWA (action level 2.5 µg/m³) as Cr(VI).** Full PPE, mist suppressant, local exhaust, and medical surveillance per OSHA 29 CFR 1910.1026 apply. This is the chrome line's responsibility — not the nickel line's. Know which chemistry runs where, and guard your nickel bath from chrome drag-back (even 1 ppm Cr⁶⁺ wrecks the nickel deposit).

• FINAL RINSE & DRY (STAGE 08)

- DI water rinse, ambient, gentle immersion 15–30 s.
- Warm air dry: 120–140°F max.
- No cure time for the nickel itself; topcoats cure per their own TDS.
- Handle with clean, dry, lint-free gloves — fingerprints on fresh nickel can stain or interfere with topcoat adhesion.

• TEACH-BACK CHECKLIST

- ☐ List at least four post-treatment paths and when each applies.
- ☐ Explain why nickel must be re-activated before a topcoat plate.
- ☐ State the Cr(VI) hazard and PEL if a chrome topcoat is used, and whose responsibility it is.
- ☐ Explain why build-up parts are plated oversize and machined back.

• TOP MISTAKES & QUICK TROUBLESHOOTING

SYMPTOM	PROBABLE CAUSE	WHAT TO DO
Nickel tarnishes/hazes in storage	Expected — bare nickel is not corrosion-proof	Anti-tarnish dip or topcoat; VCI packaging
Topcoat peels from nickel	Nickel passivated on standing; not re-activated	Re-activate with light acid dip before topcoat; minimize delay
Cracking after bake on thick deposits	Deposit was already high-stress (contamination)	Fix at plating — analyze metals, lower CD; bake relieves H, not deposit stress
Stain/fingerprints under topcoat	Bare or dirty handling	Clean lint-free gloves; keep parts clean and dry
Build-up cracks during machining	Excessive internal stress in deposit	Fix at plating (temp, pH, CD, contamination); sound plain Watts machines well



HEADS-UP

Bake ovens = burn and fire hazard — follow oven LOTO and hot-work procedures; no flammables near ovens. Anti-tarnish and seal chemistries: consult the product SDS. If a chromium topcoat is applied, full Cr(VI) controls apply. Nickel remains a skin sensitizer through all handling — gloves at all times.

SIGN-OFF – STATION 07/08

Trainee: _____ Trainer: _____ Treatment path: _____ · Dried without spots? Y / N

"I confirm I selected the correct post-treatment for the job, re-activated where a topcoat followed, handled with clean gloves, and dried the part cleanly."

PART FOUR — BACK MATTER & CERTIFICATION

— EVERY TERM, PLAIN AND SIMPLE

GLOSSARY

IF A WORD ON THE FLOOR CONFUSES YOU, IT'S IN HERE

Activation (acid dip): A short dip in dilute acid that dissolves the invisible oxide film so nickel can bond to bare metal.

Anode: The positive electrode — bars or baskets of nickel that dissolve to feed nickel into the bath.

Anode bag: A cloth bag around the anode that catches “fines” (bits of undissolved anode) so they don't roughen the deposit.

ASF (amps per square foot): Current density — how much current is spread over the area being plated. Metric version is A/dm².

ASTM B689: The standard for electroplated **engineering** nickel coatings. **Type 1** designates deposits from solutions with **no brighteners, hardeners, or stress-control additives** — the class that matches a plain Watts deposit. *(Not to be confused with ASTM B456, which covers decorative nickel-plus-chrome; ASTM B733, which covers electroless nickel, does **not** apply to this process.)*

ASTM B850: The standard for the post-plate **bake** that relieves hydrogen embrittlement in high-strength steel.

Bake (H-relief): Heating a plated part (375 ±25°F, ≥8 hr, within 4 hr of plating) to drive absorbed hydrogen out of high-strength steel.

Barrel plating: Plating many small parts at once in a rotating perforated barrel.

Boric acid: The pH buffer in the bath — holds acidity steady at the part's surface and prevents burning. Handle as a reproductive-hazard concern.

Burning: A rough, dark, powdery deposit at high-current areas — caused by too-high CD, low boric acid, low metal, low temp, or high pH.

Carbon treatment: A periodic purification that removes organics and iron from the bath using activated carbon and peroxide.

Cathode: The negative electrode — your part, where nickel deposits.

Cathode efficiency: The fraction of current that deposits metal rather than making gas. Watts runs 93–97%.

Conductivity (µS/cm): A measure of dissolved ions in rinse water — the “contamination number.” Higher = dirtier rinse.

Current density (CD): See ASF.

Deposition rate: How fast the coating grows — ~25 µm/hr at ~20 ASF; scales with current density.

Drag-in / drag-out: The film of solution a part carries out of one tank (drag-out) and into the next (drag-in) — the reason rinses exist.

Ductility: How much a metal can bend or stretch before it cracks. Plain Watts nickel is highly ductile.

Dummy plating: Plating onto a scrap cathode at low current to pull metallic contaminants out of the bath.

Electroclean: Cleaning with electrical current running through the part; gas bubbles scrub off the last microscopic film.

Electroforming: Growing nickel thick on a mandrel, then removing the mandrel to make a freestanding part.

• GLOSSARY (CONTINUED)

Hull Cell: A small angled test cell that plates a panel across a range of current densities at once — read here for *soundness*, not brightness. Standard test: 267 mL, 2 amps, 10 minutes.

Hydrogen embrittlement: Brittleness in high-strength steel caused by absorbed hydrogen; can cause sudden cracking under load. Relieved by baking.

Internal stress: Built-in tension (or compression) locked into a deposit as it grows; controlled by temperature, pH, and CD.

Ion: An atom carrying an electrical charge; nickel travels the bath as Ni^{2+} .

LOTO (Lockout/Tagout): Physically locking and tagging power off before electrical maintenance.

Nickel itch: Allergic contact dermatitis from nickel exposure — cumulative and permanent once sensitized.

Passivation (of anodes): When a non-sulfur anode goes inert and stops dissolving; prevented by adequate chloride and anode area.

Plain Watts nickel: A Watts bath run *without* a brightener system — matte to semi-lustrous, ductile, sulfur-free, engineering nickel.

Rack plating: Plating parts individually hung on a current-carrying frame.

Rectifier: The power supply that converts AC to the steady DC used for plating.

S-Rounds / sulfur-depolarized nickel: Nickel anodes with a trace of sulfur that dissolve cleanly and efficiently — the general-purpose choice.

Salvage / build-up: Rebuilding worn or undersize parts by plating thick, then machining to dimension.

Substrate: The base part being plated (steel, cast iron, copper alloy, etc.).

TDS (Technical Data Sheet): The supplier's instruction document for a product — always follow it for additive levels.

Throwing power: A bath's ability to plate evenly into recesses and low-current areas; improved by higher chloride, pH, temp, and agitation.

Water-break test: Watching water sheet off a cleaned part — an unbroken sheet means clean; beading means oil remains.

Watts bath: The foundational nickel electrolyte (nickel sulfate + nickel chloride + boric acid), characterized by O. P. Watts in 1916.



TIP

Half of “talking the language” is just these terms. When you can say “the low-CD end of the Hull panel is hazy — probably zinc or copper, let’s dummy at 2–5 ASF” and mean it, you’re communicating like a professional plater — and your chemistry supplier can actually help you over the phone.

FIELD REFERENCE

TROUBLESHOOTING
QUICK-INDEX

FIND YOUR SYMPTOM, READ THE MOST LIKELY STAGE, GO TO THE FULL GUIDE



REMEMBER

Most plain Watts defects trace back to **three root causes**: a prep failure upstream (cleaning/activation), a contaminated bath (metals or organics), or running outside the temperature/pH/CD window. Diagnose with a **Hull Cell** — read for soundness — before you start dumping chemistry.



HEADS-UP

Before changing acid strength, dip time, or anything that increases metal attack, check whether the part is **>40 HRC** — those changes raise hydrogen-embrittlement risk. When in doubt, ask your supervisor and lean on the relief bake.

IF YOU SEE...	MOST LIKELY STAGE TO CHECK	GO TO
Peeling / blistering	Cleaning & activation (prep)	Stations 01, 03
Skip plating / bare spots	Cleaning (organic film / oxide)	Station 01
Pitting (gas pits)	Low wetter / poor agitation	Station 05
Burning at edges/high CD	pH high, boric low, CD high, temp low	Station 05, Ch 5
Dull / hazy at low CD	Metallic contamination (Cu/Zn/Pb)	Ch 5.4
Roughness / nodules	Particulate / anode fines / filtration	Station 05
Cracking in thick build-up	Contamination / stress / CD / temp	Ch 5.6, Station 05
Nickel bath pH dropping	Acid drag-in from activation	Station 04
Iron / metals climbing in bath	Pre-plate rinse / spent pickle	Station 04, Ch 5.4
Anodes passivating	Low chloride / anode area / non-sulfur anodes	Ch 5.3
Topcoat peels off nickel	Nickel not re-activated	Station 07
Sudden fracture of hardened part	Missed / late H-relief bake	Stations 06, 07, Ch 5.7



TIP

Notice how many symptoms trace back *upstream* of where they show up. Most downstream defects are born upstream — a peeling deposit at Station 05 was usually made at Station 01 or 03. Fix the root, not the symptom.

— TEMPLATE — PRINT ONE PER OPERATOR

OPERATOR TRAINING MATRIX

THE TRAINER RECORDS THE DATE THE OPERATOR IS CLEARED FOR INDEPENDENT WORK AT EACH STATION



REMEMBER

An operator is "line-qualified" only when every station they run is signed. Nobody touches the **rectifier**, the **acid dip**, or **bath maintenance** solo until safety competency (PPE, SDS, LOTO, emergency response) is demonstrated. Safety qualification comes first, always.

STATION	TRAINED (DATE)	SIGN-OFF INITIALS	RE-CERT DUE
01 Soak Clean / Electroclean	_____	_____	_____
02 Rinse (Pre-Activation)	_____	_____	_____
03 Acid Activation	_____	_____	_____
04 Rinse (Pre-Plate)	_____	_____	_____
05 Watts Nickel Plating	_____	_____	_____
06 Post-Plate Rinse / Bake	_____	_____	_____
07 Post-Treatment	_____	_____	_____
08 Final Rinse / Dry	_____	_____	_____
Bath maintenance (carbon/dummy)	_____	_____	_____

Operator name: _____ Hire/start date: _____

Supervisor name: _____ Review cycle: ☐ Annual ☐ Other: _____



TIP

Re-verify each operator annually, and **immediately** after any process or chemistry change. Note the recertification date in the cell. People drift from procedure over time, and a yearly refresher catches bad habits before they become defects — or injuries.

20 QUESTIONS • PASSING SCORE 80% (16/20)

WATTS NICKEL COMPLETION TEST

COVERS EVERY STATION PLUS SAFETY — ANSWER IN YOUR OWN WORDS WHERE ASKED



REMEMBER

Passing score: 80% (16 of 20 correct). Any **safety** question answered wrong (Q5, Q6, Q14, Q19) is an automatic fail of the safety gate — re-teach and re-test before sign-off, regardless of total score. The Answer Key follows — no peeking until you're done!

Name: _____ Date: _____ Score: _____ / 20 **Safety-gate Qs (all must be correct):**

5, 6, 14, 19

Q1. Plain Watts nickel differs from bright nickel mainly because plain Watts:

- A. Uses cyanide instead of sulfate B. Runs at room temperature C. Has no organic brightener system (no carrier/brightener/leveler) D. Uses chromium anodes

Q2. (Short answer) What is the **water-break test**, and what result tells you a part is clean?

Q3. The Watts plating bath typically runs at what temperature for engineering work?

- A. Below freezing B. 130–140°F C. 200–250°F D. Room temperature only

Q4. (Short answer) Name the main job of the **acid activation** stage. In one sentence, why does the manual say “no activation, no adhesion”?

Q5. [SAFETY GATE] Which station is described as the **single greatest source of hydrogen absorption** on the line?

- A. Soak clean B. Acid activation (acid dip) C. Post-plate rinse D. Final rinse

Q6. [SAFETY GATE] A part is **> 40 HRC** (high-strength steel). Per ASTM B850, the hydrogen-embrittlement relief bake is:

- A. 200°F for 1 hour, any time within a week B. 500°F for 30 minutes C. No bake — the deposit is ductile D. 375 ±25°F for at least 8 hours, started within 4 hours of plating

Q7. (Short answer) The pre-activation rinse (Station 02) targets **< 500 µS/cm**, but the pre-plate rinse (Station 04) targets **< 200 µS/cm**. Why is the rinse closer to the nickel bath held to a *tighter* standard?

Q8. The Watts plating bath operates at what pH?

- A. 3.5–4.5 B. 11.0–13.5 C. 7.0 (neutral) D. 1.5–2.2

Q9. (Short answer) What is the job of **boric acid** in the bath, and what happens if you neglect it?

Q10. The Watts bath's cathode efficiency is about:

- A. 40–50% B. 60–80% C. 93–97% D. 100%, always

Q11. (Short answer) A part comes off the line **matte / satiny**, not shiny. What should you do, and what must you **not** do?

Q12. You run a Hull Cell on a plain Watts bath. What are you reading it for?

A. Brightness and leveling response B. Soundness — burning, roughness, pitting, contamination C. Brightener concentration D. Sulfur content

Q13. Which metallic contaminant is the **most damaging at the lowest level** (as little as 1 ppm) in the nickel bath?

A. Iron (Fe) B. Zinc (Zn) C. Hexavalent chromium (Cr⁶⁺) D. Boric acid

Q14. (Short answer) [SAFETY GATE] Name **two** nickel-specific health hazards an operator must guard against, and one control for each.

Q15. Why must the **pre-plate rinse** be especially good before the nickel bath?

A. To warm the parts up B. It isn't important C. To add brightener D. Because acid, iron, and other metals dragged in accumulate permanently in a bath that runs for years

Q16. For a plain Watts (sulfur-free) end use such as brazing preforms, which anode is preferred?

A. Non-sulfur (rolled/electrolytic) nickel, with chloride kept high B. Sulfur-depolarized S-Rounds C. Chromium anodes D. Copper anodes

Q17. (Short answer) Why does a **ductile, low-stress** plain Watts deposit **not** exempt a high-strength steel part from the bake?

Q18. Why must the Watts tank and fixtures be **PP, CPVC, or lined** rather than mild steel?

A. To save money B. The hot acidic chloride bath corrodes mild steel (adding iron and failing structurally) C. Steel is too heavy D. It doesn't matter



SAFETY SECTION — MUST BE CORRECT TO CERTIFY

Questions 5, 6, 14, and 19 cover hazards that can permanently harm you or others. A miss on any safety item means re-teach and re-test before sign-off.

Q19. (Short answer / Safety) [SAFETY GATE] List **four** pieces of PPE you must wear at the **acid activation** station, and name the procedure you must perform before any maintenance on the **rectifier**.

Q20. (Short answer) Plain Watts nickel does not level or shine like bright nickel. Name **two** honest tradeoffs (limitations) of a plain Watts deposit compared to bright nickel.

End of test. Hand to your trainer for grading.

FOR TRAINERS & SELF-CHECKERS

ANSWER KEY

SHORT-ANSWER RESPONSES DON'T NEED TO MATCH WORD-FOR-WORD — LOOK FOR THE KEY CONCEPTS IN BOLD

**HEADS-UP**

Keep this page out of the trainee's copy. Questions **5, 6, 14, and 19** are safety-critical — any wrong answer there requires re-training on that topic before sign-off, regardless of total score.

- Q1. C** — It has no organic brightener system (no carrier, primary brightener, or leveler).
- Q2.** Dip and pull the part and watch the water. If it **sheets off in an unbroken, continuous film**, the part is clean. If it **beads up**, oil remains — reclean. (It's free; use it.)
- Q3. B** — 130–140°F is the engineering target (full window 115–150°F); warmer gives lower stress and higher ductility.
- Q4.** It **dissolves the surface oxide** and produces a chemically active surface so nickel can bond. Without it, oxide blocks the bond and the plating won't stick (no adhesion) — especially fatal on thick build-up.
- Q5. B** — Acid activation (the acid dip).
- Q6. D** — 375 ±25°F for at least 8 hours, started within 4 hours of plating (ASTM B850).
- Q7.** Because tighter rinsing closer to the nickel bath limits **drag-in** of acid, dissolved **iron**, and other metals. The bath is **maintained for years**, so contaminants **accumulate permanently** and are hard to remove — the closer to the tank, the more damage drag-in does, so the standard tightens. (Full credit for naming pH protection, iron control, *or* "permanent accumulation.")
- Q8. A** — 3.5–4.5.
- Q9.** Boric acid is the **pH buffer** — it holds pH steady right at the part's surface (the cathode film) during plating. Neglect it and you get **burning** at high-current areas and inconsistent deposits.
- Q10. C** — 93–97%.
- Q11. Accept it — matte/satiny is correct for plain Watts.** Do **NOT** add brightener to "shine it up." If genuine shine is required, that is a topcoat or buffing decision made by the lead, not an additive to the tank.
- Q12. B** — Soundness (burning, roughness, pitting, contamination). There's no brightener to read for.
- Q13. C** — Hexavalent chromium (Cr⁶⁺), damaging at ~1 ppm.
- Q14.** Any two, each with a control: **skin sensitization / "nickel itch"** → gloves at every nickel station, minimize skin contact; **nickel mist (suspected carcinogen)**, **PEL 1 mg/m³** → local exhaust ventilation, respirator when ventilation is inadequate; **ingestion** → no eating/drinking/smoking on the line, wash before breaks. (Any two valid hazard + control pairs.)
- Q15. D** — Acid, iron, and other metals dragged in accumulate permanently in a bath that runs for years.
- Q16. A** — Non-sulfur (rolled/electrolytic) nickel, keeping chloride toward the high end to prevent passivation.
- Q17.** Because the ductility is a property of the **nickel deposit**, but the embrittlement is in the **steel substrate** underneath — from hydrogen absorbed during cleaning, activation, and plating. The bake drives that hydrogen out of the steel; the deposit being ductile doesn't change the steel's condition.
- Q18. B** — The hot acidic chloride bath corrodes mild steel, which would add iron to the bath and fail structurally; PP/CPVC/lined tanks resist it (plain PVC is marginal above 140°F).
- Q19. PPE (any four):** sealed chemical splash goggles (*not* safety glasses on this acid station), face shield, acid-resistant gloves, chemical apron, chemical-resistant boots, respirator if ventilation is inadequate. **Rectifier procedure: LOTO (Lockout/Tagout)** — de-energize and lock out the power before any rectifier maintenance.
- Q20.** Any two of: **no leveling** (reproduces every scratch/tool mark — prep must be right); **not decorative as-plated** (matte, not mirror); **softer / lower hardness** than bright nickel (if wear demands hardness); (also acceptable: runs hot, so higher energy cost than a room-temperature process).

MC letter distribution (Q1, 3, 5, 6, 8, 10, 12, 13, 15, 16, 18): A=2 · B=4 · C=3 · D=2 (11 multiple-choice items; 9 short-answer). Safety short-answers (Q14, Q19) are graded on concepts, not letters.

**REMEMBER**

A score of 16/20 passes — but every safety question (5, 6, 14, 19) must be correct to certify. If a trainee misses a safety item, re-teach it and re-test before sign-off. We don't gamble with people.

PRINT, FILL IN, SIGN



PLATING POSTERS

PLATING POSTERS INC · METAL FINISHING REFERENCE SERIES

CERTIFICATE OF COMPLETION
WATTS NICKEL PLATING TRAINING PROGRAM

This certifies that

OPERATOR NAME

has successfully completed the **Watts Nickel Plating** training course — covering all eight process stages (Clean, Rinse, Activate, Rinse, Plate, Rinse, Post-Treat, and Final Rinse/Dry), the plain Watts bath chemistry and maintenance, anode selection, contamination control, Hull Cell soundness diagnostics, internal stress and ductility, hydrogen-embrittlement control, troubleshooting, and the safety practices for alkaline cleaners, mineral acids, the acid nickel bath, and post-treatment — and has passed the Completion Test with a score of _____ / 20, including all required safety competencies.

This operator is hereby cleared for **independent work** at the certified stations listed on the Operator Training Matrix.

Date of completion: _____ Re-certification due: _____

TRAINEE SIGNATURE

CERTIFYING TRAINER

SUPERVISOR

Training reference only. Verify all parameters against your chemistry supplier's TDS, customer specifications, and applicable regulatory requirements before production use. Nickel compounds are classified as suspected carcinogens and known skin sensitizers; hexavalent chromium, where used in a topcoat, is a confirmed human carcinogen — always consult current SDS and comply with OSHA, EPA, REACH, and local regulations. *Regulatory data reviewed 2026-09-17: exposure limits reflect published U.S. federal OSHA PELs as of this date (soluble nickel 1 mg/m³ TWA as Ni; sulfuric acid mist 1 mg/m³ TWA; HCl 5 ppm ceiling; Cr(VI) 5 µg/m³ TWA, action level 2.5 µg/m³). Referenced standards (ASTM B689, B850, B571, F519; ISO 4526; AMS 2403/2424) should be checked against their current revision. State limits (e.g., Cal/OSHA) and ACGIH TLVs may be stricter. Boric acid: treated as a reproductive-hazard concern (no federal OSHA PEL in the confirmed set). Verify against current OSHA tables and supplier SDS before production use.*

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